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Bachelor Thesis

Optimal Exercise Strategy for Swing Options on Mean-Reverting Assets

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Contents

Abstract

This thesis computes, characterises, and analyses the optimal exercise policy for swing options on a seasonally mean-reverting asset, using backward dynamic programming on a trinomial lattice under an arithmetic Hull-White model calibrated to Henry Hub data. The optimal policy is a single price threshold, rising in cumulative volume, falling in time, and near bang-bang. Scarcity of remaining rights drives the rise in cumulative volume, decaying optionality drives the fall in time, and late in the contract the take-or-pay floor forces the threshold below the strike at volume states that would otherwise miss the floor. At the calibrated parameters the average threshold is nearly flat and converges to the strike. Mean reversion is fast enough that waiting for a better price is worth little, so errors in the estimated reversion speed barely move the policy, whereas errors in the estimated volatility shift it in proportion. Exercise nevertheless concentrates in winter, since it is the seasonal rise in the price level that lifts the spot across the flat threshold.

Keywords: swing options, optimal exercise policy, stochastic optimal control, trinomial lattice, mean reversion.

1 Introduction

Unlike a paper asset, which is bought once and held as a single position, a physical commodity is delivered and consumed as a flow over time, in amounts that vary from one delivery date to the next. This temporal, quantity-based structure introduces volume as a second state variable alongside the price. Due to the complexity of future consumption rates and limited storability of commodities, a fixed volume per delivery date is frequently uneconomic. A swing option hedges this volume risk by giving the holder the right to vary the volume taken at each delivery date, within contractual limits, while fixing the price paid through a strike K. The flexibility manages the volume risk and the fixed strike manages the price risk, in exchange for an option premium. By exercising this option, the holder can match the quantity drawn to fluctuating demand while controlling procurement costs. Swing options are widely offered by market makers and used extensively by energy companies, especially in the electricity and fossil fuel markets. They are traded both as standalone financial instruments and as structured physical transactions [1, 2, 3].

The flexibility that makes a swing option valuable also makes it difficult to price. The holder decides how much to exercise at each date, which turns valuation into a sequential decision problem. A standard European option involves no such decision and an American option involves only one. A swing option generalises the American case to many exercises, where each exercise both generates a payoff and draws down a limited budget of volume, changing which actions are admissible at every subsequent date. This coupling between today's action and tomorrow's admissible set is the source of the complexity. Because the available volume is limited, the holder must weigh the immediate payoff against saving capacity for a more favourable opportunity later. The action's effect on the state makes this a stochastic control problem rather than an optimal stopping problem. Tracking the volume exercised so far adds a third state variable, the cumulative exercised volume, to the time and price that an American option requires. The valuation itself emerges as a by-product of solving this problem, but we focus on the exercise policy: when and how much to exercise. Each exercise yields an immediate payoff, and the total payoff is the sum of these across all exercise dates. The objective is to maximise the expected discounted total payoff. We are therefore looking for the optimal exercise policy: a rule that prescribes, for any combination of time, price, and cumulative exercised volume, how to optimally act next. Solving this control problem requires a model for the dynamics of the underlying price, in particular one capturing the mean reversion and seasonality characteristic of energy commodities [1, 3].

To model mean reversion we use the Hull-White model, an extension of the Ornstein-Uhlenbeck process that makes the equilibrium a deterministic function of time, so the level the process reverts to shifts throughout the year. We take the seasonal level to be a truncated Fourier series in time, with a constant, a linear trend, and the first two harmonics of the annual cycle, which captures the recurring seasonal pattern of energy prices using few parameters [4, 5].

The research question of this thesis is: "What is the optimal strategy for exercising swing options on a seasonally mean-reverting asset?" The main contribution is computing, characterising, and analysing the optimal exercise policy. Ample research exists on pricing these options, yet the optimal exercise strategy from the holder's perspective is less thoroughly characterised, and its general structural properties remain largely open [1, 3]. The scope of this thesis is to model the underlying spot price, price the option, compute the optimal exercise policy, and analyse the policy from several perspectives. We additionally discuss the computational complexity of the method and alternative approaches to computing the optimal policy.

Pricing swing options is most commonly framed as a discrete stochastic optimal control problem. The holder chooses a quantity to exercise for every date, subject to local and global volume constraints, and the contract value is given by a backward dynamic programming equation over the underlying price and the cumulative exercised volume [3, 6, 7, 8]. In the continuous-time limit the problem becomes a Hamilton-Jacobi-Bellman equation or variational inequality, characterised through viscosity solutions [9, 10]. A separate line of work develops the rigorous existence theory through optimal multiple stopping, generalising single-stopping theory to the multiple-exercise case [11, 12]. This formulation fits contracts where each right is a single unit exercise, while the variable-volume contract studied here is the natural generalisation. The present work adopts the control formulation [7, 8], and the lattice it solves on places it in the dynamic-programming methods rather than the simulation-based methods.

In commodity literature, gas prices are typically modelled as mean-reverting and seasonal [3, 6, 13]. The classic underlying model for gas-swing valuation is the one-factor seasonal mean-reverting model, introduced for this purpose by Jaillet, Ronn and Tompaidis [3] in its geometric form, in which the logarithm of the deseasonalised price follows an Ornstein-Uhlenbeck process. We adopt the arithmetic counterpart, the Lucia-Schwartz specification [14]. Here the deseasonalised price itself is Ornstein-Uhlenbeck. Because the payoff is then affine in the lattice state, the optimal exercise boundary admits a clean threshold characterisation in that state, which is central to the structural analysis of the optimal exercise policy. The cost is that the Gaussian deviation permits negative prices, a trade-off discussed in Chapter 2.

For this class of models, both the option value and the optimal exercise strategy are obtained by backward induction over a three-dimensional state of time, price, and cumulative exercised volume [3, 6]. Additionally, this is the standard approach for swing valuation and therefore the setting adopted here [15].

The three-dimensional state and its backward-induction solution produce an exercise policy, whose structure remains only partially characterised. The coupling across dates is a consequence of the global volume constraints, specifically the take-or-pay floor: the minimum cumulative volume the holder must exercise, which forces decisions at later dates to depend on the current cumulative volume exercised. This classical structure was analysed by Thompson [16], who showed that the contract value depends on the prior path only through the remaining volume obligation, and that exercising only part of the available volume can be optimal even for a price-taking holder. Prior research regarding the exercise boundary is setting-specific. It has been characterised analytically for swing puts under geometric Brownian motion with a refracting period, where the stopping boundaries are continuous and monotone in time [17]. Additionally, value-function monotonicity has been established in continuous-time formulations [10, 11]. The optimal policy is often of bang-bang type [3, 6] but intermediate volumes can be optimal. Specifically near the take-or-pay floor, where partial exercise serves only to reduce the penalty [16], or when the global and local constraints are not commensurate, so that the integrality conditions guaranteeing bang-bang fail [7].

For the seasonal mean-reverting, variable-volume contract studied here, however, neither the boundary's structure nor its sensitivity to the model parameters has been characterised. This thesis addresses that gap and computes the exact policy over the entire discretised state space, analysing how the exercise boundary responds to the reversion speed, volatility, strike, the volume constraints, and take-or-pay floor.

The thesis proceeds as follows. We first describe the problem of finding the optimal exercise strategy for a swing option on a seasonally mean-reverting asset. We model the underlying, price the option under that model, then numerically compute the optimal exercise policy, and finally analyse it. We do not claim the proposed model fits the market exactly. We adopt it since it captures the main empirical properties of energy prices, mean reversion and seasonality, keeps the payoff linear in the state, and is standard in the energy-markets literature. Chapter 2 presents the model of the underlying asset. Chapter 3 proposes the pricing and control framework, Chapter 4 presents the numerical method, and Chapter 5 covers the model calibration. Chapter 6 contains the results and validation, Chapter 7 summarises, and Chapter 8 contains a discussion and further work. The Appendix collects supporting material omitted from the main text.

2 The Model

The model consists of two components: a stochastic process for the spot price of the underlying commodity, and the specification of the swing option contract written on it.

2.1 From Ornstein-Uhlenbeck to Hull-White

Energy commodity prices behave differently from equity prices. When the spot price of a fossil fuel such as natural gas rises far above its typical level, high prices suppress demand and draw stored gas into the market, pushing the price back down. When the price falls far below that level, the opposite occurs. Prices are therefore pulled toward an equilibrium, a property known as mean reversion. Empirical studies document this behaviour across energy markets [18, 19, 20].

The canonical continuous-time model of a mean-reverting quantity is the Ornstein-Uhlenbeck (OU) process [4], defined by the stochastic differential equation

$$dS_t = \kappa(\theta - S_t)dt + \sigma dW_t, \tag{1}$$

where $\theta \in \mathbb{R}$ is the long-run equilibrium level, $\kappa > 0$ the mean reversion speed, $\sigma > 0$ the volatility, and W_t a standard Brownian motion. The drift term pulls the process back to its mean. It is negative when $S_t > \theta$ and positive otherwise, therefore fluctuating around θ.

However, a constant equilibrium level does not describe fossil fuel markets such as natural gas and oil. Demand for these fuels is seasonal, typically peaking in winter, when heating consumption is greatest, while production is comparatively stable through the year. Since it is possible to store these fuels, the imbalance is absorbed by the storage cycle, with injection in periods of low demand and withdrawal in periods of high demand. The result is that the price level exhibits seasonality, being systematically higher in winter than in summer [21, 22, 23].

If θ were constant, every winter peak would be a temporary deviation to be corrected, instead of a recurring feature of the equilibrium. The solution is therefore to introduce a time-dependent equilibrium $\theta(t)$. This is the same idea Hull and White applied to extend the Vasicek interest-rate model [24], itself an Ornstein-Uhlenbeck process for the short interest rate, where a time-dependent mean level lets their model match an observed term structure [5]. We therefore refer to the resulting spot-price model as a Hull-White model. The identical construction appears in energy literature as the seasonal one-factor Schwartz model, or in its arithmetic form as the Lucia-Schwartz model [14, 19], and it is the standard underlying model for swing option valuation on natural gas [3, 15, 25].

2.2 The Asset Price Model

Instead of specifying $\theta(t)$ directly, we define the model through an additive decomposition of the asset price into a deterministic seasonal component and a zero-mean stochastic deviation. This is the form in which the model will be calibrated (Chapter 5) and discretised (Chapter 4), and it follows the same arithmetic specification as Lucia and Schwartz [14].

Definition 1 (Asset price process). *The spot price S_t of the underlying commodity is given by*

$$S_t = \alpha(t) + X_t, \qquad X_0 = S_0 - \alpha(0), \qquad dX_t = -\kappa X_t\, dt + \sigma\, dW_t^{\mathbb{Q}}, \tag{2}$$

where $\alpha\colon [0, T] \to \mathbb{R}$ is a deterministic, continuously differentiable function describing the seasonal price level, X_t is an Ornstein-Uhlenbeck process with zero long-run mean, and $W_t^{\mathbb{Q}}$ is a standard Brownian motion under the risk-neutral measure $\mathbb{Q}$.

This decomposition separates the two components motivated above: $\alpha(t)$ carries the deterministic seasonal pattern, while X_t captures stochastic deviations from it.

Remark 1 (Equivalence to the Hull-White model). *Definition 1 is equivalent to the classical Hull-White specification*

$$dS_t = \kappa(\theta(t) - S_t)\, dt + \sigma\, dW_t^{\mathbb{Q}}, \qquad \theta(t) = \alpha(t) + \frac{\alpha'(t)}{\kappa}. \tag{3}$$

The derivation together with the inverse relation is given in Appendix A.1. We work with α throughout, since it is the object that is estimated from data and that appears in the numerical scheme.

As a consequence of the additive form the price is Gaussian and may therefore become negative. An alternative is a geometric specification, $\ln S_t = f(t) + X_t$, which guarantees positivity.

The numerical scheme of Chapter 4 discretises the deseasonalised process X_t onto a lattice, writing X_j for the price level at spatial node j and t_i for the ith date, so that the spot at that node is $S_{i,j} = \alpha(t_i) + X_j$. The exercise boundary $S^*(i, C)$ is the lowest spot at which the holder exercises at date t_i given cumulative volume C, attained at the threshold node $j^*(i, C)$. Under the arithmetic form the per-unit payoff $S_{i,j} - K = \alpha(t_i) + X_j - K$ is affine in the lattice state X_j. That is, for a fixed date t_i it is a straight-line function of X_j, with unit slope and intercept $\alpha(t_i) - K$. Since $S_{i,j}$ is monotone in the node index j, the optimal policy admits a threshold characterisation in the lattice state (Chapter 3), and that threshold decomposes additively into the seasonal level and a deviation, $S^*(i, C) = \alpha(t_i) + X_{j^*(i,C)}$, with no nonlinear remapping between price and state.

The geometric form loses the additive form of the threshold, which then enters multiplicatively through $e^{f(t_i)+X_{j^*}}$. Then, the premium over the seasonal level is captured either by the ratio $e^{X_{j^*}}$ or by the absolute gap $e^{f(t_i)}\big(e^{X_{j^*}} - 1\big)$, which scales with the seasonal level $f(t_i)$. Neither is the single additive deviation X_{j^*}, comparable across dates and capacities, that the arithmetic form allows. The whole analysis of Chapter 6 is expressed in exactly this premium over the seasonal curve $\alpha(t)$, and reads as one interpretable quantity only because the arithmetic payoff makes it additive. A secondary, empirical reason is that Lucia and Schwartz found the arithmetic specification fit observed futures and forward prices better than the log-price version in their study of Nord Pool electricity [14].

The compromise is a Gaussian tail. At the calibrated parameters of Chapter 5 the marginal probability of a negative price is not negligible, peaking at roughly 4.6% at the seasonal trough near $t \approx 0.80$, where the seasonal level is lowest and the transient has decayed to the stationary standard deviation $\sigma/\sqrt{2\kappa} \approx 1.67$. This is a direct consequence of the large calibrated volatility, itself driven by the price-spike episodes discussed in Section 5.5. We therefore adopt the arithmetic form as the specification best suited to characterising the exercise policy, and note the geometric and jump-extended alternatives, which restore positivity at the cost of the affine structure, as directions for further work (Chapter 8).

2.3 The Seasonal Component

The seasonal component $\alpha(t)$ must capture a recurring annual pattern with a possible long-term drift, using few parameters. The two dominant deterministic specifications in energy-price literature are piecewise-constant dummy variables and Fourier functions, the latter combining sinusoidal terms with a linear trend [26]. We adopt the sinusoidal form, a standard specification for the annual seasonal level [14, 27, 28],

$$\alpha(t) = a_0 + bt + a_1 \cos(2\pi t) + b_1 \sin(2\pi t) + a_2 \cos(4\pi t) + b_2 \sin(4\pi t), \tag{4}$$

with t measured in years. The first harmonic captures the annual winter-summer cycle, the second permits asymmetry between the heating and cooling seasons. The form is smooth, periodic by construction, and linear in its coefficients, allowing estimation by ordinary least squares (Chapter 5).

The decisive reason for the sinusoidal form is smoothness. The alternatives in the literature, monthly dummy variables [14, 27] and piecewise-constant monthly factors [3], introduce jumps in α at month boundaries that the lattice's fine time grid would resolve as artificial discontinuities. The choice between the two then depends on the regularity of the seasonal pattern. Dummy variables offer more flexibility but are more sensitive to price jumps, whereas the sinusoidal form changes gradually with the season and requires fewer parameters [27, 29].

This deterministic specification is an approximation. A body of gas-specific evidence finds that seasonality in natural gas prices is in part stochastic rather than fixed, with stochastic-seasonality models outperforming deterministic ones [23, 30, 31]. We adopt the deterministic form regardless: it avoids a second stochastic factor, which would add a state dimension and enlarge the lattice. The stochastic-seasonality and jump-extended alternatives are noted as further work (Chapter 8).

2.4 Properties of the Process

The SDE for X_t in (2) admits the explicit solution

$$X_t = X_s e^{-\kappa(t-s)} + \sigma \int_s^t e^{-\kappa(t-u)}\, dW_u^{\mathbb{Q}}, \quad 0 \le s \le t, \tag{5}$$

from which the conditional mean and variance follow from the martingale property and Itô isometry (derivations in Appendix A.2):

$$\mathbb{E}_{\mathbb{Q}}[X_t \mid X_s] = X_s e^{-\kappa(t-s)}, \tag{6}$$

$$\mathrm{Var}_{\mathbb{Q}}[X_t \mid X_s] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa(t-s)}). \tag{7}$$

As $t - s \to \infty$ the conditional variance approaches the stationary variance $\sigma^2/2\kappa$, and the influence of the starting point decays exponentially at rate κ. These conditional moments determine the lattice transition probabilities in Chapter 4.

Taking expectations of the decomposition and applying (6) with $s = 0$ gives

$$\mathbb{E}\big[S_t\big] = \alpha(t) + X_0 e^{-\kappa t}, \tag{8}$$

where $X_0 = S_0 - \alpha(0)$ is the known initial deviation of the spot from the seasonal level. The expected price therefore converges from the observed spot toward the seasonal curve, with $\mathbb{E}[S_t] \to \alpha(t)$ as $t \to \infty$. Over horizons long relative to $1/\kappa$, the seasonal function is the expected price path. This identity makes the seasonal level directly readable from price data, from the forward curve under $\mathbb{Q}$ and the historical series under $\mathbb{P}$ (Chapter 5).

2.5 Measure

The asset price process of Definition 1 is stated under a risk-neutral measure $\mathbb{Q}$. This is forced by the valuation problem of Chapter 3 due to the no-arbitrage pricing principle [32]. Consequently the value function is an expectation under $\mathbb{Q}$, and for that expectation to return the arbitrage-free value, the price dynamics it averages over must themselves be the $\mathbb{Q}$-dynamics. Definition 1 is therefore stated under $\mathbb{Q}$.

The model parameters cannot be observed under $\mathbb{Q}$ directly. In Chapter 5 they are estimated from a historical record of prices, a set of observations under the physical measure $\mathbb{P}$. Estimation thus recovers parameters under $\mathbb{P}$, which must then be reconciled with the $\mathbb{Q}$-dynamics that pricing requires. The two measures are linked by Girsanov's theorem through a market price of risk λ, which we assume constant, following the standard treatment of the mean-reverting commodity model [14, 19].

This change of measure does not affect every parameter, and it is worth separating what is invariant by theorem from what is invariant only by assumption. By Girsanov's theorem the measure change alters the drift of a diffusion but leaves its diffusion coefficient unchanged [32, 33]. Therefore the volatility σ is the same under $\mathbb{P}$ and $\mathbb{Q}$ unconditionally, with nothing assumed beyond the diffusion model itself. The reversion speed κ is also preserved, but for a weaker reason that relies on the constant market-price-of-risk assumption. Under that assumption the measure change shifts the driving Brownian motion by a constant rate, $dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \lambda\, dt$, so the drift of the deviation gains only the constant term $-\sigma\lambda$. A constant added to the drift of a mean-reverting process moves the level it reverts to, not the rate at which it reverts. Therefore the deviation X_t continues to obey the Ornstein-Uhlenbeck dynamics of (2) under $\mathbb{Q}$ with the same κ and σ. Under a constant market price of risk, the seasonal level would shift by a constant, from its physical value $\alpha_{\mathbb{P}}$ to

$$\alpha_{\mathbb{Q}}(t) = \alpha_{\mathbb{P}}(t) - \frac{\sigma\lambda}{\kappa}, \tag{9}$$

derived in Appendix A.3.

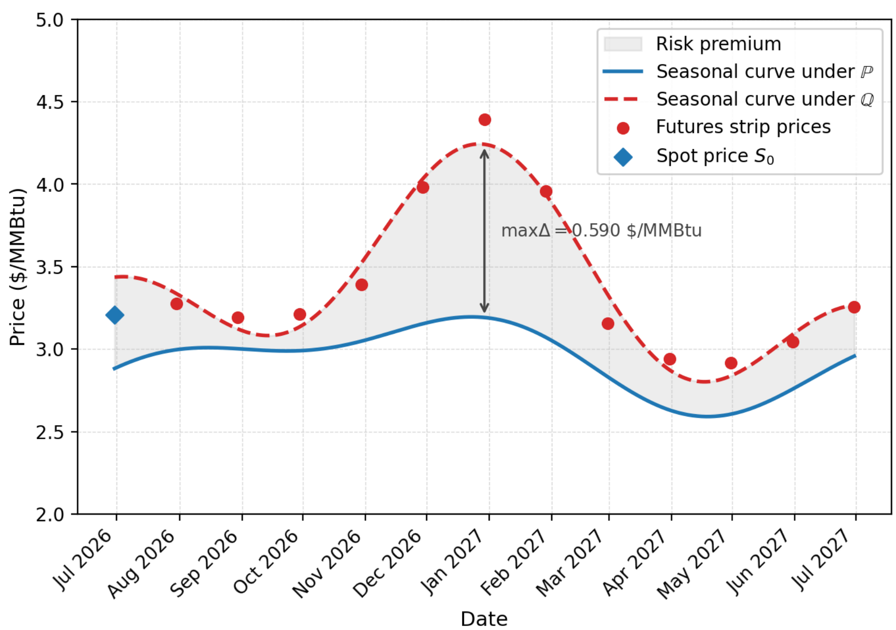

Figure 1: Fitted physical and risk-neutral seasonal curves against the forward strip.

This leaves an asymmetry that the calibration of Chapter 5 is built around. Since σ

and κ are common to both measures, they may be estimated from the historical record under $\mathbb{P}$ and used without adjustment in the risk-neutral pricing. The level is not, since under $\mathbb{Q}$ the forward price is the expected spot, which by (8) equals $\alpha_{\mathbb{Q}}(t)$ once the initial transient has decayed. The risk-neutral level is therefore read directly from the observed forward curve, $\alpha_{\mathbb{Q}}(t) = F(0,t)$, rather than by shifting $\alpha_{\mathbb{P}}$. This fits the level to the market that pricing must be consistent with, yet it discards the theoretical constant-λ relation of (9). The forward curve exhibits stronger winter seasonality than the historical shape (Figure 1), so a constant risk premium does not hold. We therefore fit $\alpha_{\mathbb{P}}$ and $\alpha_{\mathbb{Q}}$ separately and treat the constant-λ assumption as holding for the dynamics but not the level. A seasonal risk premium is left as further work (Chapter 8).

2.6 The Swing Option Contract

A swing option on S_t grants its holder the right to purchase the underlying at a fixed strike, subject to three constraints: a local (per-step) maximum, a global minimum and global maximum over the lifecycle of the contract. Let $0 = t_0 < t_1 < \cdots < t_N = T$ denote the exercise dates.

Definition 2 (Swing option contract)**.** *A swing option with strike $K > 0$ permits the holder to exercise a volume $q_i \in \{0, \ldots, q_{\max}\}$ at each date t_i, where $q_{\max} \in \mathbb{N}$ is the local volume cap. Exercising volume q_i at date t_i yields the payoff*

$$q_i \big(S_{t_i} - K\big) \tag{10}$$

which may be negative when $C_{\min}$ forces exercise near expiry, as discussed in Section 3.4. The cumulative volume is constrained by global bounds $C_{\min}, C_{\max} \in \mathbb{N}_0$ with $0 \le C_{\min} \le C_{\max}$:

$$C_{\min} \;\le\; \sum_{i=0}^{N} q_i \;\le\; C_{\max}. \tag{11}$$

A volume schedule $\mathbf{q} = (q_0, \ldots, q_N)$ satisfying these constraints, with q_i adapted, is called an admissible exercise strategy. The set of admissible strategies is denoted $\mathcal{A}$.

The global bounds distinguish the contract from a strip of independent options. The upper bound $C_{\max}$ makes the rights scarce since volume exercised today is unavailable later, so the holder withholds rights for more favourable dates. The lower bound $C_{\min}$ does the opposite, forcing exercise near expiry once the remaining capacity can no longer meet it (Section 3.4). The local cap $q_{\max}$ limits how fast the budget is consumed, so the rights cannot all be exercised at a favourable date. Tracking the cumulative exercised volume alongside time and price is what gives the valuation problem of Chapter 3 its three-dimensional state space.

3 The Pricing and Control Framework

Chapter 2 specified the asset-price dynamics, whereas this chapter prices the swing option written on them. Its value is the expected discounted payoff earned under the optimal admissible exercise strategy, so pricing the option and finding its optimal policy are the same problem [3]. We frame the contract as a discrete stochastic control problem, in which the holder chooses an exercise volume at each date subject to the local and global limits of Definition 2.

First we define the per-date payoff and the set of admissible strategies. Then we define the value function over the resulting state space and give the backward recursion that solves it. Next we examine the case in which a minimum-volume obligation forces exercise at a loss. Finally we extract the optimal exercise policy. Throughout, valuation is carried out under the risk-neutral measure $\mathbb{Q}$.

3.1 Payoff and Admissible Strategies

A swing option gives the holder the right to a sequence of exercise decisions, one for every exercise date t_i. We partition $[0, T]$ into N subintervals of length Δt.

$$t_i = i\Delta t, \qquad i = 0, \dots, N, \qquad \Delta t = \tfrac{T}{N}. \tag{12}$$

The payoff for a single exercise is the difference between the spot price and the strike.

Definition 3 (Per-unit exercise payoff)**.** *Let $K > 0$ be the strike. The payoff from exercising a single unit at spot price S is*

$$\xi(S) = S - K. \tag{13}$$

Thus the payoff from exercising a volume $q \in \mathbb{N}_0$ at price S is $q\xi(S) = q(S - K)$, which may be negative.

Justification for using the signed payoff $S - K$ rather than the truncated $(S - K)^+$ is given in Section 3.4.

The holder's decision at each date is the volume for the given time step, bounded by the local cap.

Definition 4 (Exercise action and local constraint)**.** *At each date t_i the holder selects a volume $q_i \in \{0, \dots, q_{\max}\}$, where $q_{\max} \in \mathbb{N}$ is the maximum allowed local (per-step) exercisable volume.*

Whether a given q is admissible depends on the constraints (11). That is, the cumulative volume may not exceed the global maximum $C_{\max}$ and enough exercise opportunities must remain for the global minimum $C_{\min}$ to be reachable. This second requirement is the source of the forced-exercise behaviour studied in Section 3.4. We formalise this by defining the feasible action set $\mathcal{Q}_i(C)$, the subset of $\{0, \dots, q_{\max}\}$ containing exactly the volumes that, at time step i with cumulative volume C, satisfy all of the constraints above.

Definition 5 (Feasible action set)**.** *Let $C \in \{0, \dots, C_{\max}\}$ denote the cumulative volume exercised strictly before date t_i. The set of feasible actions at (i, C) is*

$$\mathcal{Q}_i(C) = \big\{q \in \{0, \dots, q_{\max}\} : \underline{q}_i(C) \le q \le \overline{q}_i(C)\big\}, \tag{14}$$

where the upper and lower bounds are

$$\overline{q}_i(C) = \min\big\{q_{\max},\ C_{\max} - C\big\}, \tag{15}$$

$$\underline{q}_i(C) = \max\big\{0,\ C_{\min} - C - (N - i)q_{\max}\big\}. \tag{16}$$

The lower bound $\underline{q}_i(C)$ is the forced minimum exercised at t_i (Section 3.4).

An admissible exercise strategy is the combination of such actions, one per date, each chosen based on the information available at that time step.

Definition 6 (Admissible exercise strategy)**.** *Write* $C_i = \sum_{k=0}^{i-1} q_k$ *for the cumulative volume exercised strictly before date* t_i *(so* $C_0 = 0$ *and* $C_{i+1} = C_i + q_i$*). A strategy* $\mathbf{q} = (q_0, \dots, q_N)$ *is admissible if it is adapted to the filtration generated by* $(S_{t_i})_{i=0}^{N}$ *and satisfies the global bounds of Definition 2, equivalently,* $q_i \in \mathcal{Q}_i(C_i)$*, for all* i*.*

3.2 State Space and the Value Function

To numerically value the option we discretise both time and the underlying asset price. At every date t_i (12) we represent the asset price without its seasonal component on a spatial lattice indexed by j, such that the asset price at node (i, j) is

$$S_{i,j} = \alpha(t_i) + X_j, \qquad X_j = (j - m)\,\Delta X, \qquad j = 0, \dots, 2m, \tag{17}$$

where the X_j are nodes of the deseasonalised process $X_t = S_t - \alpha(t)$, defined in Definition 1. Additionally, we track the cumulative exercised volume C, which determines the number of exercises left. The discrete state of the contract is therefore the triple (i, j, C).

This control problem is a finite-horizon Markov decision process, so the value at any date depends solely on the current state, not on the path that led to it. Two properties support this. First, the asset is Markovian: X_t is an Ornstein-Uhlenbeck process, the solution of an Itô SDE and therefore Markov [33, 34]. Second, both the reward and the constraints are functions of the current state alone. The per-date payoff $q\xi(S)$ depends only on the spot price and the volume chosen, since the strike is fixed, and the feasible action set $\mathcal{Q}_i(C)$ depends only on the date index, the cumulative exercised volume, and the fixed contract parameters, as established in Section 3.1. The optimal expected discounted reward over the remaining dates is therefore a function of (i, S_{t_i}, C), which allows a value function defined on the triple.

At the initial state, time is zero ($i = 0$), the exercised volume is zero ($C = 0$), and the spot S_0 is known.

Definition 7 (Swing option value)**.** *With interest rate* $r \in \mathbb{R}^+$*, the time-zero value of the swing option is*

$$V_0 = \max_{\mathbf{q} \in \mathcal{A}} \mathbb{E}_{\mathbb{Q}} \left[\sum_{i=0}^{N} e^{-rt_i} q_i \xi(S_{t_i}) \right], \tag{18}$$

where $\mathbb{Q}$ *is the risk-neutral measure and* $\mathcal{A}$ *the admissible strategies of Definitions 2 and 6.*

An identical construction determines the value at an arbitrary date. The state (i, S, C) records that the holder has reached date t_i with spot S and cumulative exercised volume C, and $V_i(S, C)$ is the largest expected discounted payoff obtainable by exercising optimally over the remaining dates. Relative to (18), the sum now runs over $k = i, \dots, N$, discounting is measured from t_i through the factor $e^{-r(t_k - t_i)}$, and the expectation conditions on the current state.

Let $\mathcal{A}_i(C)$ be the set of admissible continuation strategies $(q_i, \dots, q_N)$ adapted to the price information with $q_k \in \mathcal{Q}_k(C_k)$ for every $k \geq i$, where $C_i = C$ and $C_{k+1} = C_k + q_k$.

Writing $\mathbb{E}_{\mathbb{Q}}^{i,S,C}[\cdot] = \mathbb{E}_{\mathbb{Q}}[\cdot \mid S_{t_i} = S,\ C_i = C]$ for the conditional expectation, the value at state (i, S, C) is

$$V_i(S,C) = \max_{\mathbf{q}\in\mathcal{A}_i(C)} \mathbb{E}_{\mathbb{Q}}^{i,S,C}\left[\sum_{k=i}^{N} e^{-r(t_k-t_i)} q_k \xi(S_{t_k})\right]. \tag{19}$$

Because $\mathcal{A}_0(0) = \mathcal{A}$ and $S_{t_0} = S_0$ is known, evaluating (19) at $i = 0$ returns (18), so the contract value is $V_0 = V_0(S_0, 0)$.

3.3 The Bellman Recursion

Equation (19) is defined as the value function of a maximisation over all continuation strategies, which cannot be evaluated directly. However, the Markov structure established in Section 3.2 solves this. The price is Markov and the reward at each date depends only on the current state, so the value at (i, S, C) separates into the best admissible immediate reward plus the discounted value of acting optimally from the next state. This dynamic programming principle reduces the global maximisation problem to a sequence of one-step problems solved by backward induction. The terminal date is solved directly, and each earlier date follows once the values at the next date are known.

At expiry the current action does not affect any future payoff, so the terminal value reduces to the highest feasible immediate reward,

$$V_N(S,C) = \max_{q\in\mathcal{Q}_N(C)} q\xi(S) = \max_{q\in\mathcal{Q}_N(C)} q(S-K). \tag{20}$$

When $0 \in \mathcal{Q}_N(C)$ the holder can decline an unfavourable asset price by choosing $q = 0$, so the maximum is non-negative and the per-unit payoff behaves as the truncated $(S - K)^+$. If instead $0 \notin \mathcal{Q}_N(C)$ the holder must exercise and may realise a loss, a case treated in Section 3.4.

At earlier dates, the value function combines the immediate reward of the current action with the discounted expectation of the value one step ahead, where $S' = S_{t_{i+1}}$ is the price at the next date,

$$V_i(S,C) = \max_{q\in\mathcal{Q}_i(C)} \Big\{ q(S-K) + e^{-r\Delta t}\, \mathbb{E}_{\mathbb{Q}}^{i,S,C}\big[V_{i+1}(S', C+q)\big] \Big\} \tag{21}$$

for $i = N-1, \ldots, 0$. Recursively applying (21) computes the value function at every state, recovering the contract value at the initial node as $V_0(S_0, 0)$. The maximiser at each state is the optimal action. Collecting these over the state space defines the optimal policy from Section 3.5.

The conditional expectation is evaluated on the trinomial lattice. The current state is the node (i, j) with $S = S_{i,j}$, where the price moves to one of three successor nodes at t_{i+1}. The continuation value at the current node is the value function at the successor nodes weighted by the transition probabilities.

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}\big[V_{i+1}(S', C+q) \mid (i,j)\big] &= p_u^{(j)}\, V_{i+1}\big(S_{i+1,j_u}, C+q\big) \\ &+ p_m^{(j)}\, V_{i+1}\big(S_{i+1,j_m}, C+q\big) + p_d^{(j)}\, V_{i+1}\big(S_{i+1,j_d}, C+q\big)\end{aligned} \tag{22}$$

The trinomial lattice of Chapter 4 provides the transition probabilities $p_u^{(j)}, p_m^{(j)}, p_d^{(j)}$ and the successor indices j_u, j_m, j_d, and establishes boundary conditions.

The recursion is consistent with the value function of Section 3.2: the one-step factor $e^{-r\Delta t}$ compounds over successive steps to the multi-step factor $e^{-r(t_k - t_i)}$ of (19). The following principle records this [3, 35, 36], with a proof by backward induction on the Markov decomposition of the reward given in Appendix B.1.

Proposition 1 (Dynamic programming principle)**.** *The value function defined in* (19) *satisfies the terminal condition* (20) *and the recursion* (21)*, and* $V_0(S_0, 0)$ *coincides with the time-zero value* V_0 *of* (18)*.*

3.4 Forced Exercise and the Payoff Below the Strike

The two global bounds of Definition 2 play asymmetric roles. The upper bound $C_{\max}$ limits opportunity by restricting how much the holder may take, but never forces action. The lower bound $C_{\min}$ is an obligation. If the holder has exercised too little and few dates remain, the local capacity left over those dates may be insufficient to reach $C_{\min}$ unless exercise begins now, regardless of profitability. The holder is then compelled to exercise even when the spot lies below the strike, accepting a loss to meet the contract. This is the take-or-pay feature of physical gas supply contracts, in which the buyer commits to a minimum offtake over the delivery period.

The lower bound $\underline{q}_i(C)$ of (16) makes this obligation precise. After a volume q is taken at t_i, the $N-i$ remaining dates each contribute at most $q_{\max}$, so the largest cumulative volume still attainable by expiry is $C + q + (N-i)\, q_{\max}$. The minimum-take constraint requires this to be at least $C_{\min}$, which gives $q \geq C_{\min} - C - (N-i)q_{\max}$. Together with $q \geq 0$ this is exactly $\underline{q}_i(C)$, the smallest volume the holder may take at t_i.

Forced exercise is the case where this lower bound is strictly positive. The condition

$$\underline{q}_i(C) > 0 \quad \Longleftrightarrow \quad C + (N-i)\, q_{\max} < C_{\min} \tag{23}$$

makes $q = 0$ infeasible, so $0 \notin \mathcal{Q}_i(C)$ and the holder must take at least $\underline{q}_i(C)$ units.

This distinction fixes the correct form of the payoff. When $0 \in \mathcal{Q}_i(C)$ the holder exercises only when profitable and never takes a unit below the strike, so the effective per-unit payoff is the truncated $(S-K)^+$. When $0 \notin \mathcal{Q}_i(C)$ the holder has no such freedom and may be compelled to exercise while $S < K$, making the realised per-unit payoff $S - K$ negative. Modelling the payoff as $(S-K)^+$ everywhere would erase these forced losses, overstate the contract value, and distort the exercise policy in the forced region. We therefore use the signed quantity of Definition 3 as the per-unit payoff and let the feasible set $\mathcal{Q}_i(C)$, rather than a truncation, decide when the holder may decline. The truncation then arises only where it should, as the outcome of the maximisation at states where $q = 0$ is admissible.

The effect on the optimal policy is examined in the results chapter, where the exercise boundary is pushed below the strike in the forced region.

Remark 2 (Relation to the implementation)**.** *The recursion realises this distinction with no special-casing of the payoff. Maximising the signed reward* $q(S-K)$ *over* $\mathcal{Q}_i(C)$ *in* (21) *covers both regimes, so in code it is enough to construct* $\mathcal{Q}_i(C)$ *correctly. The payoff is always the signed* $S - K$*.*

3.5 Extracting the Optimal Exercise Policy

The value function and the optimal policy answer different questions. The value function gives what the contract is worth while the policy gives how the holder should act. The optimal action at a state is the maximiser of the same Bellman objective whose maximum defines $V_i(S, C)$,

$$q_i^*(S,C) \in \arg\max_{q\in\mathcal{Q}_i(C)} \Big\{ q(S-K) + e^{-r\Delta t}\mathbb{E}_{\mathbb{Q}}^{i,S,C}\left[V_{i+1}(S', C+q)\right] \Big\}. \tag{24}$$

Extracting the policy requires no extra computation. As the backward induction of Section 3.3 evaluates (21) at each state, the maximising volume is recorded alongside the value, so the entire policy over (i, j, C) is produced in the same pass. It is defined at every date, including expiry, where it is the maximiser of the terminal condition (20).

Write $\pi^*(i,j,C) = q_i^*(S_{i,j}, C)$ for the resulting policy on the lattice. It is exact and defined at every node, not only along a single realised trajectory. This is a direct advantage over least-squares Monte Carlo [37], which recovers the policy only along simulated paths rather than over the full state space, and makes the policy analysis tractable.

Remark 3 (Tie-breaking)**.** *The maximiser in (24) is not always unique. Write $\mathcal{M}_i(S, C)$ for the nonempty set of maximisers in (24). When $|\mathcal{M}_i(S,C)| > 1$ we resolve the tie by taking the smallest optimal volume,*

$$q_i^*(S,C) = \min \mathcal{M}_i(S,C),$$

so that $\pi^*(i,j,C) = q_i^*(S_{i,j}, C)$ is single-valued at every node. Such ties arise only where two actions deliver exactly equal value, so the convention affects isolated states, not regions. Fixing it removes the ambiguity that the exercise-boundary and monotonicity diagnostics of Chapter 6 would otherwise inherit. The choice matches the implementation, which keeps the first maximiser encountered as q increases from zero, and that first maximiser is exactly $\min \mathcal{M}_i(S, C)$*.*

4 Numerical Method

Chapters 2 and 3 posed the valuation problem in continuous time and continuous price. Neither the conditional expectation in the Bellman recursion (21) nor the value function itself admits a closed form once the contract constraints of Section 3.1 are imposed, so the recursion must be solved numerically. This requires discretising both the price process and time, replacing the continuous state space with a finite lattice on which the backward induction can be carried out.

This chapter develops that discretisation. We first construct the lattice on the deseasonalised process X_t, fixing the time step Δt and price step ΔX, and derive the transition probabilities by moment-matching against the conditional moments of (6)-(7). We then describe the backward sweep that implements the recursion of Section 3.3 on the lattice, recovering both the option value and the optimal exercise policy. We then sharpen the resolution of the exercise boundary, first by sub-stepping the lattice between exercise dates, then by interpolating the threshold within a grid cell. We close with the computational complexity of the resulting algorithm.

4.1 Trinomial Lattice

The trinomial lattice method is used to approximate the stochastic component of the asset price and solve the backward recursion of Section 3.3 on it. It consists of two parts: an underlying grid that discretises time and price, and a recombining trinomial tree, sitting on the grid, that approximates the dynamics of X_t. Since the seasonal component $\alpha(t)$ is deterministic, the lattice represents only the zero-mean deviation X_t, not a process with drift, so its scale is set by the dynamics alone.

In option pricing, it is more common to use the binomial tree of Cox, Ross and Rubinstein [38] where price moves to one of two successor nodes at each step. We use a trinomial tree, with three successors, for two reasons. First, for a fixed number of steps the trinomial scheme converges faster and attains greater accuracy than the binomial one [39]. The second, more fundamental, reason follows from the mean reversion of X_t. When the spatial step ΔX is fixed by the convergence requirement of Section 4.2, each node of a binomial tree has a single free probability, since the two branch probabilities sum to one. With one free probability a binomial node cannot match both the conditional mean and variance of the increment. Matching even the mean alone forces the probability outside $[0, 1]$ at nodes far from equilibrium, where the restoring drift is large. A proof is given in Appendix C.3. A trinomial node has two free probabilities which match the first two conditional moments of X_t with ΔX fixed. The third branch keeps every probability non-negative across the grid. At extreme nodes the branching is redirected inward to preserve this validity (Section 4.3), following the Hull-White tree construction [40] as applied to energy swing options by Kluge [25]. We now define the lattice formally.

Definition 8 (Trinomial lattice). *Fix a number of time steps $N \in \mathbb{N}$ and a half-width $m \in \mathbb{N}$. The trinomial lattice is the grid of nodes*

$$(t_i, X_j), \quad t_i = i\Delta t, \quad \Delta t = \tfrac{T}{N}, \quad i = 0, \dots, N, \tag{25}$$

$$X_j = (j - m)\Delta X, \quad j = 0, \dots, 2m, \tag{26}$$

where Δt is the time step and ΔX the price step. The space index j is centred at $j = m$, so that $X_m = 0$ and the grid is symmetric about the long-run level of the deseasonalised process X_t, spanning $2m + 1$ values from $-m\,\Delta X$ to $m\,\Delta X$.

On this grid, X_t is approximated by a recombining trinomial tree. From an interior node (t_i, X_j) the process moves at t_{i+1} to one of the three successor nodes

$$X_j \mapsto \begin{cases} X_{j+1} = X_j + \Delta X & \text{w.p. } p_u^{(j)}, \\ X_j = X_j & \text{w.p. } p_m^{(j)}, \\ X_{j-1} = X_j - \Delta X & \text{w.p. } p_d^{(j)}, \end{cases} \tag{27}$$

with $p_u^{(j)} + p_m^{(j)} + p_d^{(j)} = 1$. The tree recombines, so the node count at step i grows linearly in i rather than as 3^i. The transition probabilities $p_u^{(j)}, p_m^{(j)}, p_d^{(j)}$ depend on the node through the conditional moments of X_t and are derived in Section 4.3. At the top and bottom nodes the move set is modified to keep the tree within the grid, as detailed in that section.

4.2 Discretisation

The lattice of Definition 8 discretises two dimensions, time and price. Time reuses the exercise-date partition of (12). The price dimension requires more care.

We discretise the deseasonalised process X_t, following Section 3.2. Working with X_t rather than S_t keeps the grid compact. The deviation is stationary, fluctuating about zero with stationary variance $\sigma^2/(2\kappa)$, the long-run limit of the conditional variance in (7), so a single grid of fixed width covers its range over the entire horizon. The spot price has a level that moves with the season through $\alpha(t)$, and a grid spanning S_t directly would have to be far wider to track that level across the year, with most of its nodes carrying negligible probability at any given date.

The spatial step is

$$\Delta X = \sigma\sqrt{3\,\Delta t}, \tag{28}$$

the standard spacing for a trinomial tree [41]. It ties the step to the per-step standard deviation of the process, so that the moment-matched transition probabilities of Section 4.3 stay valid and non-negative.

It remains to fix the half-width m. X_t is unbounded, yet a grid covering its full reachable range over N steps is unnecessary. Since X_t reverts to zero, its stationary law is Gaussian and concentrated about the mean, so distant nodes carry vanishing probability and the grid can be truncated well within that range. Under the stationary distribution, with standard deviation $\sigma/\sqrt{2\kappa}$, the mass beyond n standard deviations is $2\big(1-\Phi(n)\big)$, where Φ is the standard normal distribution function. At $n = 4$ this is about 6.3×10^{-5}, so truncating at four standard deviations,

$$X \in \left[-\frac{4\sigma}{\sqrt{2\kappa}}, \frac{4\sigma}{\sqrt{2\kappa}}\right], \tag{29}$$

discards a negligible share of the stationary mass and keeps the truncation error small.

Beyond the stationary spread, the grid must also contain the initial deviation from the seasonal level $X_0 = S_0 - \alpha(0)$. X_0 can lie beyond four stationary standard deviations, and the grid must reach it irrespective of the argument above. The two requirements are independent, and m must satisfy both. With

$$m_\sigma = \left\lceil \frac{4\sigma/\sqrt{2\kappa}}{\Delta X} \right\rceil \quad \text{and} \quad m_{X_0} = \left\lceil \frac{|X_0|}{\Delta X} \right\rceil \tag{30}$$

denoting the half-widths that cover the stationary spread and reach the start, we set

$$m = \max\big(m_\sigma,\, m_{X_0}\big) + 2. \tag{31}$$

The buffer of two nodes keeps the top and bottom of the grid, where the branching is modified in Section 4.3, a little beyond the region the process reaches in practice, so the boundary adjustment does not disturb the active part of the lattice.

4.3 Transition Probabilities

The lattice defined in Definition 8 is completed once each node has the corresponding transition probabilities attached. We determine them by matching the two conditional moments of the one-step increment of X_t to the moments of the Ornstein-Uhlenbeck process.

Two features distinguish this construction. First, the moments we match are exact conditional moments of the process, taken from the closed-form transition law of Chapter 2, and not the moments of a time-discretised approximation of the SDE. For example, an Euler-Maruyama step would treat the increment as Gaussian with mean $-\kappa X_j \Delta t$ and variance $\sigma^2 \Delta t$, which agree with the true moments only to first order in Δt [42]. Matching the exact moments instead reproduces the mean reversion and the conditional variance of the process at every step, however coarse Δt is. Second, the scheme is probabilistic, which implies that the discretised process is a Markov chain on the lattice whose one-step law is $\big(p_u^{(j)}, p_m^{(j)}, p_d^{(j)}\big)$. An explicit finite-difference discretisation of the associated pricing equation would compute the same continuation value, and is in fact formally equivalent to a trinomial tree. However, we work in the tree form because its probabilities make the moment matching, the validity conditions, and the boundary handling explicit.

Conditioned on the lattice value $X_{t_i} = X_j$, the one-step increment $X_{t_{i+1}} - X_{t_i}$ has mean and variance

$$M = -X_j\big(1 - e^{-\kappa \Delta t}\big), \qquad V = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa \Delta t}\big), \tag{32}$$

obtained from the conditional mean (6) and variance (7) with $t - s = \Delta t$. The variance V is the same at every node, so the probabilities vary across the lattice only through the drift M, which is proportional to X_j.

From an interior node the increment takes the values $\{+\Delta X,\, 0,\, -\Delta X\}$ with probabilities $p_u^{(j)}, p_m^{(j)}, p_d^{(j)}$. These must sum to one, reproduce the first moment M and the second raw moment $V + M^2$ of the increment. This gives the system

$$p_u^{(j)} + p_m^{(j)} + p_d^{(j)} = 1, \tag{33}$$

$$\big(p_u^{(j)} - p_d^{(j)}\big)\, \Delta X = M, \tag{34}$$

$$\big(p_u^{(j)} + p_d^{(j)}\big)\, \Delta X^2 = V + M^2. \tag{35}$$

Solving the system (Appendix C.1) gives

$$p_u^{(j)} = A + B, \qquad p_m^{(j)} = 1 - 2A, \qquad p_d^{(j)} = A - B, \tag{36}$$

where $A = \frac{V+M^2}{2\Delta X^2}$ and $B = \frac{M}{2\Delta X}$.

The spacing $\Delta X = \sigma\sqrt{3\Delta t}$ of (28) keeps all probabilities in $[0, 1]$. The binding constraint is interior non-negativity $p_m^{(j)} \geq 0$, that is $\Delta X^2 \geq V + M^2$. For small Δt, $V \approx \sigma^2 \Delta t$ and $M = O(\Delta t)$, so $V + M^2 \approx \sigma^2 \Delta t$ and $\Delta X^2 = 3\sigma^2 \Delta t$ gives $p_m^{(j)} \approx \frac{2}{3}$, with $p_u^{(j)} = p_d^{(j)} \approx \frac{1}{6}$ in the driftless case, all strictly inside $(0, 1)$. Drift enters $p_m^{(j)}$ only through the M^2 term, an $O(\Delta t)$ correction to the leading $\frac{2}{3}$, and to leading order shifts $\pm B$ between $p_u^{(j)}$ and $p_d^{(j)}$, so the factor of three in ΔX determines the margin. A smaller step $\Delta X^2 = \sigma^2 \Delta t$ would force $p_m^{(j)} \approx 0$ which leaves no room for drift. A much larger one would push $p_m^{(j)}$ toward one and waste resolution. At the top and bottom of the truncated grid the symmetric move set would step off the lattice, so the branching is redirected inward. At the top node the three successors are the current node and the two below it, and the increment takes the values

$\{0,\, -\Delta X,\, -2\Delta X\}$. Matching the two moments gives (derivation in Appendix C.2).

$$p_u^{(j)} = 1 + A + 3B, \qquad p_m^{(j)} = -2A - 4B, \qquad p_d^{(j)} = A + B. \tag{37}$$

At the bottom node, with increments $\{0,\, +\Delta X,\, +2\Delta X\}$,

$$p_u^{(j)} = A - B, \qquad p_m^{(j)} = -2A + 4B, \qquad p_d^{(j)} = 1 + A - 3B. \tag{38}$$

Both boundary forms still sum to one. The A and B coefficients each sum to zero across the three branches, and the lone 1 is carried by the stationary branch. The buffer of two nodes built into the half-width in (31) keeps these boundary rows beyond the region the process reaches in practice, so the asymmetric branching preserves the moments at the edge of the grid without affecting the interior, where most of the probability mass sits.

At the one-step-per-date resolution of this section the formulae above are non-negative throughout the region the chain visits, so the guard is inactive. Under the sub-stepping of Section 4.5 the redirected branching of (37) and (38) can instead produce a negative middle probability once Δt is fine, since the per-step reversion $\kappa\Delta t \approx 0.00116$ at $K_{\text{sub}} = 100$ is then too weak to justify one-sided branching. The implementation clips any negative value to zero and renormalises the triple to sum to one. This guard is inactive up to $K_{\text{sub}} = 8$ and first triggers between $K_{\text{sub}} = 8$ and 16. At the production resolution it fires only at the two extreme rows of the lattice (2 of 101 columns, most negative raw probability -0.220 at the bottom row, $X = -6.96$), which lie just beyond four stationary standard deviations and are reached with negligible probability under the priced dynamics. Repricing with the guard disabled changes the option value by 0.172 (1.09×10^{-4} in relative terms), two orders of magnitude below the discretisation error, so the clip does not affect the reported results. The probabilities and their successor nodes are what the lattice expectation (22) requires. The successor indices (j_u, j_m, j_d) are $(j{+}1,\, j,\, j{-}1)$ at an interior node, $(j,\, j{-}1,\, j{-}2)$ at the top, and $(j{+}2,\, j{+}1,\, j)$ at the bottom, which is the information the backward sweep of the next section uses to evaluate the recursion.

4.4 The Backward Sweep

Together the Bellman recursion of Section 3.3 and the lattice expectation (22) determine the value at every state. This subsection states the routine applied to the lattice, filling in the value function and recording the optimal action across the whole state space in a single backward pass.

During the sweep two arrays over the discrete state (i, j, C) are maintained: the value $V_{i,j,C}$ and the optimal action $\pi^*(i, j, C)$, both shaped $(N+1)\times(2m+1)\times(C_{\max}+1)$. The pass starts at expiry, $i = N$. For every node j and capacity C,

$$V_{N,j,C} = \max_{q\in\mathcal{Q}_N(C)} q(S_{N,j} - K), \qquad \pi^*(N, j, C) \in \arg\max_{q\in\mathcal{Q}_N(C)} q(S_{N,j} - K), \tag{39}$$

the terminal value (20) evaluated at each lattice node.

For an earlier date $i < N$ the value at (i, j, C) is the best candidate over the feasible actions, where each candidate is the immediate reward plus the discounted continuation read from the values already computed at step $i+1$,

$$V_{i,j,C} = \max_{q\in\mathcal{Q}_i(C)} \Big\{ q(S_{i,j} - K) + e^{-r\Delta t}\, \mathbb{E}_{\mathbb{Q}}^{i,S,C}\big[V_{i+1}(S', C{+}q)\big] \Big\}, \tag{40}$$

with $\pi^*(i,j,C)$ the maximising q. The successor indices (j_u, j_m, j_d) and probabilities $p_u^{(j)}, p_m^{(j)}, p_d^{(j)}$ come from Section 4.3, including the modified sets at the top and bottom rows that impose the boundary conditions. Because the action increments capacity from C to $C+q$, the continuation is read at $C+q$ rather than C, which is why the sweep carries and loops over C.

The traversal order follows from (40), since the update at step i reads only values at step $i+1$. Sweeping i from $N-1$ down to 0, and within each i over all j and C, therefore visits every state only after the states it depends on are filled, so one backward pass completes the arrays exactly, without the repeated fixed-point iteration required for infinite-horizon problems.

Algorithm 1 Backward sweep

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1. **Initialise.** For each $j \in \{0, \dots, 2m\}$ and $C \in \{0, \dots, C_{\max}\}$, set $V_{N,j,C}$ and $\pi^*(N,j,C)$ by (39).
2. **Recurse.** For $i = N-1$ down to 0, and each j and C, set $V_{i,j,C}$ and $\pi^*(i,j,C)$ by (40), breaking ties by the smallest maximiser.
3. **Read off.** Return the contract value $V_0 = V_{0,j_0,0}$, where j_0 is the node with X_{j_0} nearest $X_0 = S_0 - \alpha(0)$, together with π^* over all (i,j,C).

The output is the pair the rest of the thesis uses. The number $V_{0,j_0,0}$ is the contract value $V_0(S_0, 0)$ of (19), and the array π^* is the optimal policy over the entire discretised state space, exact at every node, which Chapter 6 analyses.

4.5 Sub-Stepped Lattice Refinement

The spacing (28) ties the price step to the time step, $\Delta X = \sigma\sqrt{3\Delta t}$, and with one lattice step per exercise date this gives $\Delta X \approx 1.39$ USD/MMBtu at the calibrated parameters. Chapter 6 shows the voluntary exercise boundary sits within one such step of the strike, so the grid is too coarse to resolve the object of study. Refining by raising N directly is not feasible since the exercise dates are contract terms, and adding lattice steps as exercise dates would create a different contract with more exercise opportunities.

We therefore decouple the lattice clock from the exercise clock. Fix $K_{\text{sub}} \in \mathbb{N}$ and run the lattice on $N_{\text{f}} = K_{\text{sub}} N$ steps of length $\Delta t_{\text{f}} = T/N_{\text{f}}$, so the price step shrinks to

$$\Delta X_{\text{f}} = \sigma\sqrt{3\,\Delta t_{\text{f}}} = \frac{\Delta X}{\sqrt{K_{\text{sub}}}}. \tag{41}$$

Exercise is permitted only on the original dates, the lattice indices with $i \bmod K_{\text{sub}} = 0$. For other indices the feasible set is $\{0\}$ and the update reduces to the discounted continuation. This mirrors how Bermudan options are valued numerically, where the price process evolves on a lattice finer than the exercise schedule, and exercise is admitted only on the contract dates. The admissibility bounds of Definition 5 count remaining exercise dates, not lattice steps. With $d(i) = \lfloor \frac{(N_{\text{f}}-i)}{K_{\text{sub}}} \rfloor$ dates remaining, the forced minimum becomes $\underline{q}_i(C) = \max\{0,\, C_{\min} - C - d(i)q_{\max}\}$. The transition probabilities are the moment-matched triple of Section 4.3 evaluated at Δt_{f}, and the per-step discount factor $e^{-r\Delta t_{\mathrm{f}}}$ compounds over K_{sub} sub-steps to the per-date factor. The contract, its volume bounds, and its exercise opportunities remain unchanged, so only the resolution of the price process improves. At $K_{\mathrm{sub}} = 1$ the construction reduces to the lattice of Definition 8 exactly, and the contract value moves by less than half a percent from $K_{\mathrm{sub}} = 1$ to 100, confirming the refinement leaves the pricing essentially unchanged while it sharpens the boundary.

4.6 Extracting the Exercise Boundary

The backward sweep returns the policy π^* on the grid, so the lowest exercising node overstates the true threshold by up to one price step since the boundary lies somewhere between the last waiting node and the first exercising node. We recover its position within the cell from the value function. Define the exercise advantage of one unit at node (i, j) with cumulative volume C,

$$g(i, j, C) = (S_{i,j} - K) + e^{-r\Delta t_{\mathrm{f}}}\, \mathbb{E}_{\mathbb{Q}}^{i,j}\big[V_{i+1}(S', C{+}1) - V_{i+1}(S', C)\big], \tag{42}$$

the immediate payoff of the unit less the continuation value of keeping it, with the expectation computed via (22). Exercise of the unit is favourable exactly where $g > 0$, and under the single-threshold structure verified in Section 6.2, g changes sign once across the reachable nodes. The boundary $S^*(i, C)$, the lowest spot price at which the holder exercises at date t_i with cumulative volume C, is exactly the zero of g, located by linear interpolation between the bracketing nodes $j^*{-}1$ and j^*. Writing $g^- = g(i, j^*{-}1, C)$ and $g^+ = g(i, j^*, C)$ for the bracketing values,

$$S^*(i, C) = \alpha(t_i) + X_{j^*-1} + \Delta X_{\mathrm{f}} \frac{-g^-}{g^+ - g^-}. \tag{43}$$

Since g is a difference of value functions it is smooth across one cell, so the linear zero is accurate well below ΔX_{f}. Where no sign change brackets the threshold, in the forced region and where every reachable node exercises, no voluntary boundary exists and none is reported. The terminal date is likewise omitted, since without a continuation the advantage degenerates to the payoff. All boundaries in Chapter 6 use this extraction on the sub-stepped lattice.

4.7 Computational Complexity

The computational complexity of the backward sweep follows from Algorithm 1. On the unrefined lattice ($K_{\mathrm{sub}} = 1$) the recursion visits N time steps and at each step it iterates over the $2m + 1$ lattice nodes and $C_{\max} + 1$ capacity levels, maximising over the $q_{\max} + 1$ feasible actions. Each action evaluates a three-branch continuation in constant time. The running time is therefore

$$O\big(N m C_{\max} q_{\max}\big). \tag{44}$$

The half-width m is not independent of N. By (31) and (28), $m = O\big(1/\Delta X\big) = O\big(1/\sqrt{\Delta t}\big) = O\big(\sqrt{N}\big)$, since $\Delta X = \sigma\sqrt{3\Delta t}$ and $\Delta t = T/N$. Substituting into (44), the running time in N is $O\big(N^{3/2} C_{\max} q_{\max}\big)$, superlinear rather than linear because refining the time grid also widens the price grid.

The sub-stepping of Section 4.5 scales the cost the same way. The fine lattice has $N_\text{f} = K_\text{sub}N$ steps and, by (41), half-width $m_\text{f} = O\big(\sqrt{K_\text{sub}N}\big)$. The N exercise dates each maximise over $q_\text{max} + 1$ actions, while the $(K_\text{sub} - 1)N$ sub-step indices admit only $q = 0$ and cost a single continuation evaluation, giving

$$O\big(N^{3/2}C_\text{max}\big(\sqrt{K_\text{sub}}\, q_\text{max} + K_\text{sub}^{3/2}\big)\big). \tag{45}$$

For fixed q_max the refinement therefore multiplies the running time by $K_\text{sub}^{3/2}$ where the step count contributes a factor K_sub and the widened grid a further $\sqrt{K_\text{sub}}$.

The sweep stores the value and policy arrays over the full fine state space, each of shape $(N_\text{f} + 1) \times (2m_\text{f} + 1) \times (C_\text{max} + 1)$, giving space complexity $O\big(N_\text{f} m_\text{f} C_\text{max}\big) = O\big(K_\text{sub}^{3/2} N^{3/2} C_\text{max}\big)$. Unlike a price-only valuation, which can discard past time slices, the policy π^* over all (i, j, C) is itself the deliverable and must be retained in full.

Setting $C_\text{min} = 0$ and $C_\text{max} = q_\text{max} = 1$ recovers the single-exercise American case, at cost $O\big(N^{3/2}\big)$. The capacity dimension, sized by C_max, is the computational factor the swing contract adds over the American option. The lower bound C_min constrains the feasible actions but does not affect this cost.

5 Calibration

We established the asset price model, the valuation framework, and the numerical scheme in Chapters 2 to 4, leaving parameters as free symbols. However, to compute the option values and exercise policies for the results chapter, we require real values. Therefore we calibrate the model to natural gas using historical and forward Henry Hub data such that the resulting value and policy describe a real contract. We also resolve the open question from Section 2.5, on how to turn estimates from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$ which pricing requires.

In the model we estimate three parameters: the seasonal level $\alpha(t)$, mean-reversion speed κ and the volatility σ. Due to the measure asymmetry, we must estimate them in two distinct ways. Volatility and reversion speed remain identical under the physical and risk-neutral measures, σ by Girsanov and κ under the constant market-price-of-risk assumption. We therefore estimate both from historical spot time series data, which is under the physical measure $\mathbb{P}$, and use them unchanged in pricing. The seasonal level differs between the two measures by the risk premium. The level under the risk-neutral measure is equivalent to the expected spot price under $\mathbb{Q}$, the value the forward curve quotes. We therefore read it straight from forward prices as $\alpha_\mathbb{Q}(t) = F(0, t)$.

We start with the two datasets, then estimate $\alpha_\mathbb{Q}(t)$, κ and σ. We then record the contract parameters and the remaining model inputs, which are specified rather than estimated. We close with the calibration results and with the diagnostics: the seasonal regression quality, residual seasonality, and the probability of negative prices.

5.1 Data

Both datasets are sourced from the Henry Hub market, so they refer to the same delivery point. The valuation-date spot close gives S_0 and the historical series the $\mathbb{P}$-dynamics. We price the contract with a one-year horizon $T = 1$.

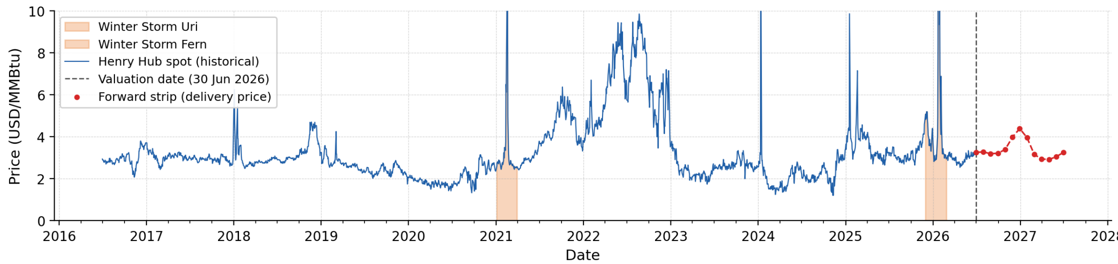


Figure 2: Henry Hub daily spot (30th June 2016-30th June 2026) with the appended forward strip.

The Henry Hub is the standard delivery point and benchmark for NYMEX natural gas futures, the principal reference for the North American gas market [43, 44]. Its price is the financial baseline for the industry and therefore this thesis. Henry Hub-linked pricing has also extended into European and Asian markets [45, 46], and it is standard in energy-derivatives literature [47].

The first dataset is the EIA daily Henry Hub spot price in USD/MMBtu [48] retrieved on 1st July 2026. It records one price per trading day, the price for immediate delivery, and is a plain time series indexed by the observation date. We fix the valuation date t_0 at 30th June 2026 and take the ten years ending there, from 30th June 2016 to 30th June 2026, giving 2,512 daily observations. Justification for the length of the window is given in Section 5.3.

The second dataset is the Henry Hub futures strip, the NYMEX settlement prices for the Henry Hub natural gas contract in USD/MMBtu [49], recorded on the valuation date t_0. It is a snapshot, rather than a time series. At t_0 it gives the prices of monthly contracts delivering across the year ahead, spanning many maturities at a single date. The spot and futures strip are complementary: the spot varies the observation date at a fixed, immediate delivery horizon, whereas the strip varies the delivery date at a fixed observation date. Because of that relationship, the spot carries the $\mathbb{P}$-dynamics and the strip the $\mathbb{Q}$-level. We take the twelve monthly contracts delivering in $(t_0, t_0 + T]$, from which we determine the seasonal level $\alpha_{\mathbb{Q}}$.

We measure time t in fractional years on the trading-day clock of Section 4 so that consecutive trading observations count as one step of Δt, and non-trading days and missing data points are dropped rather than filled. Since the dropped days fall under the same per-step approximation already applied to weekends, isolated gaps leave the estimates essentially unchanged. Both datasets are quoted in USD/MMBtu, dollars per one million British thermal units.

Two winter storms produce short, extreme spikes in the series (Figure 2). In February 2021 Winter Storm Uri cut supply across the central United States [50], and in January 2026 Winter Storm Fern drove Henry Hub to an all-time high of $30.72/MMBtu [51]. We keep both, since they are market outcomes the volatility estimate should reflect. Their effect is analysed in Section 5.5.

5.2 Computing the Seasonal Curve

Under the risk-neutral measure $\mathbb{Q}$ the seasonal level is the expected spot price, and the forward curve is exactly that. We therefore read the level from the forward market rather than history, justified by no-arbitrage.

A forward struck at the delivery price $F(0,T)$ costs nothing to enter, so its value at $t=0$, the discounted risk-neutral expectation of the payoff $S_T - F(0,T)$, is zero,

$$0 = e^{-rT}\mathbb{E}_{\mathbb{Q}}\big[S_T - F(0,T)\big]. \tag{46}$$

Because $F(0,T)$ is fixed at $t=0$, it leaves the expectation, so the forward price equals the risk-neutral expected spot at delivery,

$$F(0,T) = \mathbb{E}_{\mathbb{Q}}\big[S_T\big]. \tag{47}$$

We use NYMEX futures rather than forwards. Under a constant short rate r the futures price and the forward price coincide, each equal to $\mathbb{E}_{\mathbb{Q}}[S_T]$ [32], so the strip is the right object.

Connecting this to the model closes the argument. Taking the risk-neutral expectation of $S_t = \alpha_{\mathbb{Q}}(t) + X_t$ reproduces the conditional mean (8), $\mathbb{E}_{\mathbb{Q}}[S_t] = \alpha_{\mathbb{Q}}(t) + X_0 e^{-\kappa t}$. Identifying the level with the forward curve, after the transient decays, gives

$$\alpha_{\mathbb{Q}}(t) = F(0,t), \tag{48}$$

which is the object we estimate. The initial deviation is then $X_0 = S_0 - \alpha_{\mathbb{Q}}(0) = -0.23$, small relative to the stationary spread $\sigma/\sqrt{2\kappa}$. Its effect on the expected price decays at rate κ through the transient $X_0 e^{-\kappa t}$, so $\mathbb{E}_{\mathbb{Q}}[S_t]$ matches the forward curve up to that vanishing term.

The forward strip prices only the twelve monthly maturities, but the lattice evaluates $\alpha_{\mathbb{Q}}$ at every step t_i, so we fit the seasonal form (4) of the model by ordinary least squares. With twelve points and six parameters the system is overdetermined, so the fitted curve smooths the quotes rather than passing through them, as shown in Figure 3. The fitted $\alpha_{\mathbb{Q}}$ is the seasonal level passed to the lattice.

This risk-neutral curve is distinct from the physical seasonal level $\alpha_{\mathbb{P}}$. The two differ by the risk premium of Section 2.5, and $\alpha_{\mathbb{P}}$ is fitted separately in the parameter estimation of Section 5.3.

5.3 Computing the Model Parameters

We estimate κ and σ from the historical spot series under the physical measure $\mathbb{P}$. Both parameters are common to $\mathbb{P}$ and $\mathbb{Q}$, as established in Section 2.5, so the estimates carry over to the risk-neutral pricing without adjustment. The estimation runs in two steps: we first remove the seasonal level to expose the deviation process, then fit its dynamics. The seasonal level used here, $\alpha_{\mathbb{P}}$, serves only to recover the deviation and is discarded afterward.

Fitting the seasonal form (4) to the historical spot series by ordinary least squares gives the physical level $\alpha_{\mathbb{P}}$, and subtracting it returns the deviation,

$$X_t = S_t - \alpha_{\mathbb{P}}(t). \tag{49}$$

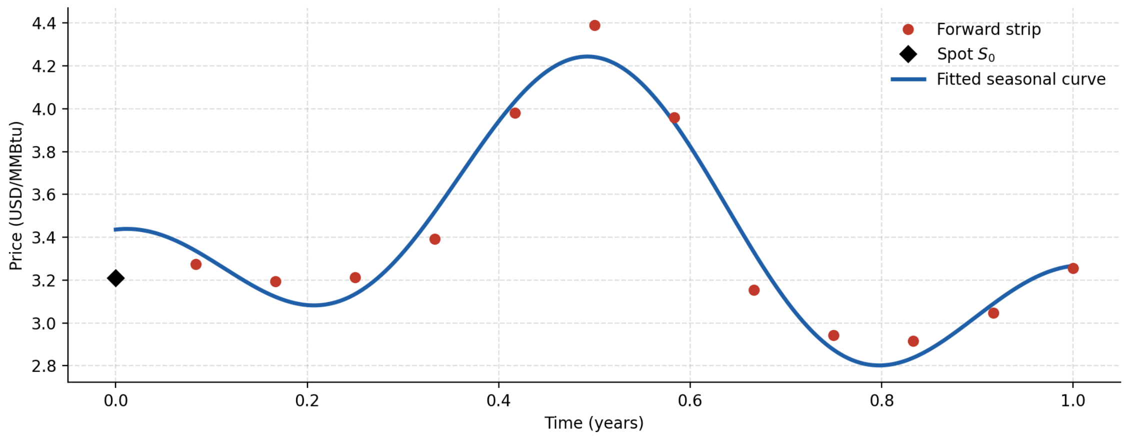


Figure 3: Fitted seasonal curve under $\mathbb{Q}$.

This X_t is the empirical counterpart of the deviation process of Definition 1.

The deviation is an Ornstein-Uhlenbeck process, so we estimate its parameters from the exact one-step law rather than an approximation. From the explicit solution (5), the deviation one step ahead is the current deviation scaled by a constant, plus a mean-zero shock,

$$X_{t+\Delta t} = e^{-\kappa \Delta t} X_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}\Big(0,\ \tfrac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa \Delta t}\big)\Big), \tag{50}$$

with the shock variance taken from (7) at $t - s = \Delta t$. This is the form of a linear regression of $X_{t+\Delta t}$ on X_t,

$$X_{t+\Delta t} = \beta_1 X_t + \varepsilon_t, \qquad \beta_1 = e^{-\kappa \Delta t}, \tag{51}$$

with no intercept. Deseasonalising leaves X_t with mean zero, so the fitted line passes through the origin. We fit it by ordinary least squares on the trading-day clock, $\Delta t = 1/252$.

The slope and the residuals carry the two parameters. The slope is the one-step reversion factor and inverting it gives

$$\kappa = -\frac{\ln \beta_1}{\Delta t}. \tag{52}$$

The spread of the points about the fitted line measures the size of the shock, and matching its variance to that of ε_t in (50) gives

$$\sigma = \sqrt{\frac{2\kappa \ \mathrm{Var}(\varepsilon_t)}{1 - e^{-2\kappa \Delta t}}}. \tag{53}$$

The one-step law (50) is the exact Ornstein-Uhlenbeck transition, the same law the lattice moment-matches in Chapter 4, so the calibrated parameters and the numerical scheme rest on the same dynamics.

The estimation leaves one choice that the data does not settle on its own: the length of the sample window. This is a bias-variance trade-off. A long window supplies more data and lowers the variance of the estimates, but it risks spanning more than one market regime, so the estimate may not represent the market today. A short, recent window is more representative of the current regime and of the period being priced, at the cost of noisier estimates. We adopt a ten-year window as a standard compromise and test its sensitivity in Section 5.5.

We analyse this tradeoff by re-running the full estimation, the deseasonalising fit together with the regression, over two families of windows. A fixed-width window rolled through the sample shows how $\hat{\kappa}$ and $\hat{\sigma}$ have moved over time and exposes any regime changes. An expanding window anchored at the valuation date t_0, with its start pushed progressively further back, shows how far into the past the estimates reach, before the older data begins to change their values. Diagnostics are reported in Section 5.5.

5.4 Contract and Determined Inputs

This section records the inputs that are specified rather than estimated. They fall into two groups. The determined model inputs, the interest rate r, the initial spot S_0, and the time grid of Section 5.1, are read directly off the market or fixed by the discretisation. On the contrary, the contract terms K, $q_{\max}$, $C_{\min}$, and $C_{\max}$ define the swing option and are chosen to represent a standard gas contract.

We take the interest rate r as the one-year Treasury constant-maturity yield on the valuation date, 3.98 percent [52], rounded to $r = 0.04$ and held constant over the life of the contract. A constant, deterministic r is the assumption already used in Section 5.2. The initial spot S_0 is the observed Henry Hub spot at t_0, read from the series of Section 5.1. The valuation date is $t_0 = 30$ June 2026, the horizon $T = 1$, and the number of steps $N = 252$.

The contract terms specify the swing option of Definition 2. Volumes are expressed in normalised units so that the payoff is linear in volume. The unit itself is an arbitrary label, and what matters for the policy is the moneyness of the strike, the ratios among the volume bounds, and the number of admissible exercise levels. We take the strike K at the money, set to the average of the fitted forward curve $\alpha_{\mathbb{Q}}$ over $[0, T]$, which is $K = a_0 + \frac{b}{2} = 3.40$.

The local cap $q_{\max}$ is the most the holder may take at a single date. Its absolute size is part of the normalisation above, but the number of admissible levels $\{0, \ldots, q_{\max}\}$ is not. It sets how finely the holder can vary the offtake, and so how much room there is for intermediate, non-bang-bang exercise. Since bang-bang behaviour is one of the policy features we study, $q_{\max}$ must be large enough to show the granularity of exercising, so we take $q_{\max} = 10$. The upper bound must satisfy $C_{\max} < N q_{\max}$, otherwise the holder could take $q_{\max}$ at every date which nullifies the constraint, and the contract would decouple into N independent single-date options. We set $C_{\max} = \frac{1}{2} N q_{\max} = 1260$, so the holder may take at most half of what taking the cap at every date would allow and must manage a limited budget. This flexibility is comparable to the swing in real gas supply contracts, where the maximum daily offtake typically ranges from just above the average up to 1.5-2 times it in high-swing contracts [53].

The lower bound $C_{\min}$ is the take-or-pay floor, the minimum total the holder must take or else pay for. It must be positive, or there is no obligation and the forced exercise of Section 3.4 never arises. We set it in the industry take-or-pay range of roughly 70 to 90 percent of $C_{\max}$ [54], taking $C_{\min} = 1050$, about 83 percent. A floor this high makes the obligation relevant, so forced exercise near expiry is an active feature of the baseline policy. This contract structure, local and global volume constraints with a take-or-pay minimum, follows the standard specification for natural gas swing options [3, 16].

Symbol	Description	Baseline	Source
$\alpha_{\mathbb{Q}}(t)$	seasonal level	fitted	§5.2
κ	reversion speed	29.2	§5.3
σ	volatility	12.75	§5.3
r	risk-free rate	0.04	[52]
S_0	initial spot	3.21	§5.1
K	strike (ATM)	3.40	specified
$q_{\max}$	local volume cap	10	specified
$C_{\max}$	global maximum	1260	specified
$C_{\min}$	global minimum	1050	specified
T	horizon	1	specified
N	time steps	252	specified

Table 1: Baseline parameter set for the calibrated swing option.

5.5 Results and Diagnostics

This section reports the calibration results and the diagnostics deferred from Sections 5.2 and 5.3, together with the negative-price check promised in Chapter 2.

Seasonal Fits

Fitting the seasonal form (4) to the twelve futures settlements of Section 5.1 gives the risk-neutral level

$$\alpha_{\mathbb{Q}}(t) = 3.485 - 0.169t - 0.445\cos 2\pi t + 0.088\sin 2\pi t + 0.396\cos 4\pi t + 0.011\sin 4\pi t, \quad (54)$$

explaining 95.8% of the variation across the strip (R^2) with a root-mean-square error of 0.09 USD/MMBtu (Figure 3). The quality of this fit matters because the lattice reads $\alpha_{\mathbb{Q}}$ at every step, therefore error in the curve is error in the seasonal level on which the entire policy is built. The first harmonic carries an amplitude of 0.45 USD/MMBtu and the second 0.40, setting a winter peak near \$4.2 against a trough of 2.80 at $t \approx 0.80$, matching the shape of the strip.

The deseasonalising fit of Section 5.3 gives the physical level

$$\alpha_{\mathbb{P}}(t) = 2.895 + 0.075\,t - 0.136\cos 2\pi t + 0.198\sin 2\pi t + 0.123\cos 4\pi t + 0.059\sin 4\pi t, \quad (55)$$

explaining 2.8% of the variation in the spot series (R^2), intentionally low. The diffusion process carries the spot variance, and the seasonal level is meant to explain only the recurring annual component. The physical level sits below the risk-neutral one with a milder seasonal swing, explained by the risk premium of Section 2.5.

Estimated dynamics

These two estimates set the dynamics of the policy problem: κ fixes how long a favourable deviation survives, and σ how large deviations get, so together they determine the value of waiting. The lag regression (51) on the $n = 2511$ formed from the 2,512 daily observations of Section 5.1 gives $\hat{\beta}_1 = 0.890$, hence

$$\hat{\kappa} = 29.2 \text{ yr}^{-1} \quad (2.6), \qquad \hat{\sigma} = 12.75 \text{ USD/MMBtu/yr}^{1/2} \quad (0.18), \tag{56}$$

with standard errors in parentheses. They are nominal, not corrected for the residual autocorrelation reported below, so they understate the true uncertainty in $\hat{\kappa}$. The reversion is fast, given that the implied half-life $\ln 2/\hat{\kappa}$ is about nine calendar days, so spikes correct within days.

Residual diagnostics

The estimates above are only as good as the one-step law (50) they assume. We therefore do two checks to probe its residuals. First, no annual pattern should survive the deseasonalisation. Regressing the residuals on the Fourier basis of (4) returns $R^2 = 0.000$, so the seasonal fit removed what it was meant to. Second, the residuals should carry no autocorrelation beyond the AR(1). The sample autocorrelations are small (Figure 4, largest about -0.09 at lag two), but with $n = 2511$ even minor departures are detectable, and a Ljung-Box test at lag 10 returns $p \approx 10^{-9}$, formally rejecting whiteness. The autocorrelation is too small to matter for the policy, but points to a second, faster-reverting factor as a refinement of the one-factor specification (Chapter 8).

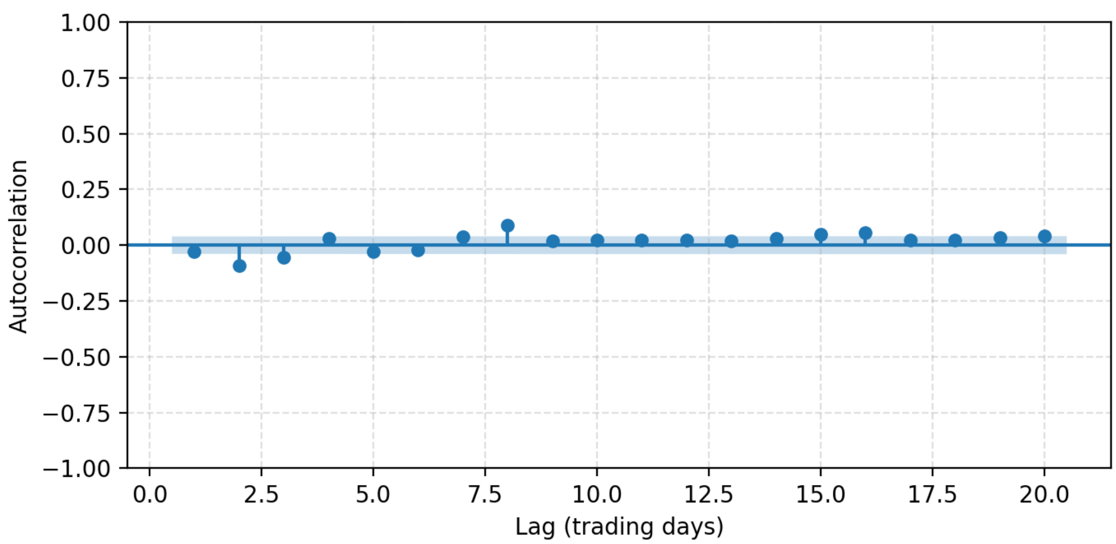


Figure 4: Sample autocorrelation of the AR(1) residuals with the 95% white-noise band.

Window Stability

The ten-year window of Section 5.1 was a choice. This check tests whether the estimates depend on it by re-running the estimation over rolling five-year windows and over expanding windows anchored at t_0 (Figure 5). Rolling estimates vary widely ($\hat{\kappa}$ from 1.4 to 84.7, $\hat{\sigma}$ from 2.2 to 27.8), and both rise as the window approaches t_0 and captures a larger share of the two winter-storm spikes of Section 5.1 (Uri, February 2021, and Fern, January 2026). Neither parameter exhibits a stable plateau, so the ten-year window is not defended as a regime-homogeneous sample but adopted as a standard choice balancing representativeness against estimation variance. The robustness rerun below isolates the spikes' contribution directly.

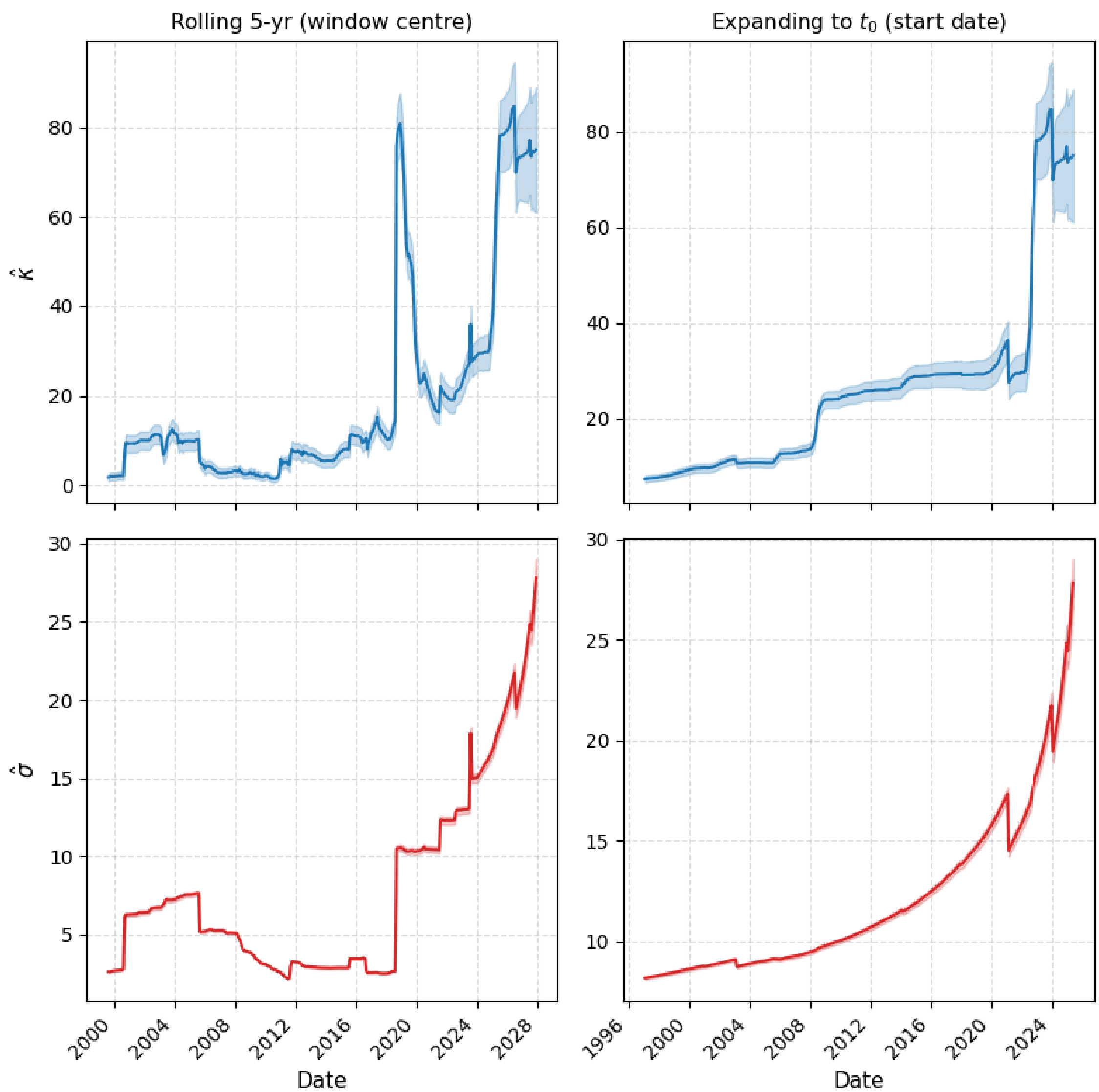


Figure 5: Rolling five-year and expanding-window estimates of $\hat{\kappa}$ and $\hat{\sigma}$.

Robustness to the spike episodes

The spike episodes of Section 5.1 enter the residual variance directly and revert within days, so both estimates depend on how those episodes are treated. If a handful of weeks drive the calibration, the baseline describes the anomalous storms rather than the market. Table 2 re-estimates the dynamics with the calendar months of Winter Storm Uri (February 2021) and Winter Storm Fern (January 2026) excluded, dropping one-step pairs that span the removed periods. Both estimates roughly halve: $\hat{\kappa}$ falls from 29.2 to 10.5 and $\hat{\sigma}$ from 12.75 to 6.46. The reversion speed is therefore not a stable feature of the series but is driven substantially by the rapid reversion of the two spikes, which a single factor reads as fast mean reversion. We retain the spikes in the baseline, since they are genuine market outcomes the policy should price against, but the sensitivity marks $\hat{\kappa}$ as the least robust input and reinforces the second-factor limitation of the residual diagnostics.

	Full sample	Excl. spikes	Change
$\hat{\kappa}$	29.2	10.5	-18.8
$\hat{\sigma}$	12.75	6.46	-6.30

Table 2: Dynamics re-estimated with the Winter Storm Uri (February 2021) and Winter Storm Fern (January 2026) episodes excluded.

Probability of negative prices

The arithmetic specification was adopted in Chapter 2 at the cost of admitting negative prices. This check quantifies that probability at the calibrated parameters and verifies the 4.6% figure quoted there. Since $S_t = \alpha_{\mathbb{Q}}(t) + X_t$ is Gaussian with mean $\alpha_{\mathbb{Q}}(t) + X_0 e^{-\kappa t}$ and standard deviation approaching the stationary $\sigma/\sqrt{2\kappa}$,

$$\mathbb{P}(S_t < 0) = \Phi\left(-\frac{\alpha_{\mathbb{Q}}(t) + X_0 e^{-\kappa t}}{\mathrm{sd}(X_t)}\right), \tag{57}$$

largest at the seasonal trough, where $\alpha_{\mathbb{Q}}$ is lowest and the transient has decayed. With $\sigma/\sqrt{2\kappa} = 1.67$ and $\alpha_{\min} = 2.80$ this gives $\Phi(-1.68) = 4.6\%$, verifying the figure quoted in Chapter 2. The probability is a consequence of the large calibrated volatility. Under the spike-excluded estimates of Table 2 it falls to 2.4%. The arithmetic specification is adequate away from the spike-driven regime but inaccurate within it, motivating the geometric and jump-extended alternatives of Chapter 8.

Table 1 in Section 5.4 collects the resulting baseline and the results chapter varies these inputs one at a time.

6 Results and Validation

This chapter computes the optimal exercise policy and analyses its structure. All results use the baseline parameter set of Table 1 unless specified otherwise. We first present the baseline policy and the exercise boundary it induces, then verify that the boundary's structural features are properties of the optimal policy rather than artefacts of a single figure, then trace how the boundary responds to the model parameters, and finally examine where exercise falls across the year. We close by comparing the optimal policy to naive alternatives, which quantifies the value of solving the control problem exactly, and validating the dynamic program against Monte Carlo simulation.

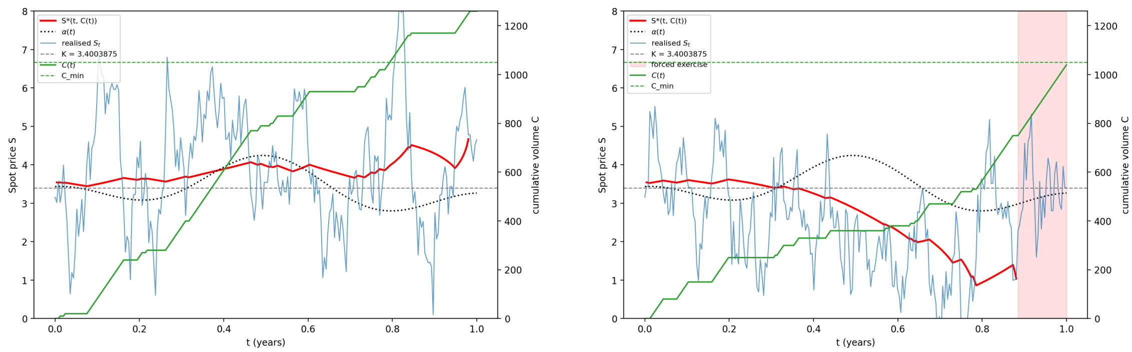
6.1 The Baseline Optimal Policy

The dynamic program returns the optimal volume q^* at every state, and Section 6.2 verifies that the exercise decision reduces to a single threshold in the spot price at every reachable state. The policy is therefore summarised by the exercise boundary $S^*(t, C)$, the lowest spot at which the holder exercises given cumulative volume C at date t. Fixing one argument gives a slice. Cumulative volume only rises, so a holder climbs a capacity trajectory rather than sitting at a fixed C. This section describes the exercise boundary surface at the calibrated parameters of Table 1.

All boundaries in this chapter are computed on the sub-stepped lattice of Section 4.5 with 100 lattice steps per exercise day, reducing the price step ΔX from ≈ 1.39 USD/MMBtu to ≈ 0.139 USD/MMBtu, and extracted at sub-node resolution by interpolating the zero of the exercise advantage (42). The contract is unchanged and the price moves by 0.05%, yet the refinement is essential for characterising the boundary. If read directly off the coarse grid, the threshold snaps to node values and overstates its height above the strike by up to a full price step. The terminal date is omitted throughout, since no continuation exists at expiry and the advantage degenerates to the payoff, leaving no threshold to define. The mean-path boundary sits at roughly \$3.54 for the first tenth of the year and holds near \$3.5 until the final month, a mean of \$3.51 across the voluntary dates, about \$0.11 above the strike, and it does not follow the seasonal curve beneath it even as $\alpha(t)$ swings from \$2.80 to \$4.25.

Figure 6 traces $S^*(t, C(t))$ along two simulated paths, each using its own exercised volume, so on these panels the comparison of spot against threshold is exactly the decision rule the holder follows. The left panel shows a path that spends most of the year in the money. Exercise comes early and often, with $C(t)$ crossing the floor by $t \approx 0.78$ and reaching $C_{\max}$ slightly before expiry. Each period of exercise lifts the threshold the holder subsequently faces, producing the sawtooth that tracks the steps in $C(t)$, and once the floor is met scarcity is the only force left. It intensifies towards expiry, where few rights and little time remain. The threshold peaks near \$4.5 at $t \approx 0.84$, falls back to about \$3.9 as exercise pauses, then climbs to \$4.7 at the final exercise date, the decline occurring as time advances at fixed cumulative volume, and the rise occurring as exercise increases that volume. When $C(t)$ reaches $C_{\max}$ the line stops, because with no rights left, no boundary exists. The right panel shows the opposite regime. The spot sits mostly below the strike, voluntary exercise is less common, and $C(t)$ accumulates too slowly to reach the floor on its own. The unmet obligation pulls the threshold below the strike. The boundary reaches the strike by $t \approx 0.33$, far ahead of the mean path, and falls to a minimum near \$0.9 at $t \approx 0.79$. The burst of exercise that follows lifts it back toward \$1.3, each unit taken easing the obligation and raising the threshold for the next, the same composition of slice-switch and within-slice movement as the sawtooth of the left panel, here running below the strike. The full-rate climb in $C(t)$ begins before the compulsion does, at $t \approx 0.85$, while exercise is still voluntary under a threshold that has fallen well beneath the strike, and from $t \approx 0.88$ the forced region imposes what the policy was already choosing, at a loss when the spot is below the strike. $C(t)$ reaches $C_{\min}$ exactly at expiry. Section 6.2 shows this is monotonicity in C. A lower cumulative volume means a nearer obligation, and a nearer obligation means a lower threshold.

Sample paths deviate substantially from the average. Figure 7 shows the representative



(a) In-the-money sample path.

(b) Out-the-money sample path.

Figure 6: The exercise boundary $S^*(t, C(t))$ along each path's own capacity trajectory.

picture. It traces the boundary along the mean exercised path $\bar{C}(t)$, the cumulative volume at each date averaged across 250,000 paths simulated under the optimal policy, so $\bar{C}(t)$ is the capacity trajectory of a representative holder and $S^*(t, \bar{C}(t))$ is the threshold that holder meets. The result is nearly flat. The boundary starts about \$0.14 above the strike, declines gently to mid-year, rises modestly from $t \approx 0.65$ to a peak near \$3.65 at $t \approx 0.89$, the date the mean path crosses the floor, and settles onto K from $t \approx 0.94$. The seasonal level swings from \$2.80 to \$4.25 around the threshold. The premium over $\alpha(t)$ consequently changes sign with the season, positive in the shoulder troughs and negative at the winter peak. The mean path meets the floor at $t \approx 0.89$ and stays clear of the forced frontier throughout, so the average holder is never forced to exercise. The date at which the boundary locks onto the strike is the first date at which the remaining rights can no longer reach $C_{\max}$, so from that point the ceiling cannot bind and the scarcity premium disappears.

This flat threshold is a symptom of the fast-reversion regime. A high volatility alone would incentivise waiting since $\sigma = 12.75$ generates large deviations, and a holder might hope to sell into them. What prices the waiting option, however, is not the size of a spike but instead its persistence. A deviation decays with a half-life of days (Chapter 5), so the spot is confined to a stationary band of width $\sigma/\sqrt{2\kappa} \approx 1.67$ around the seasonal curve, and a spike is gone before the daily cap $q_{\max}$ allows meaningful volume to be taken. Timing optionality is therefore nearly worthless and the voluntary threshold collapses toward the strike. What remains is a budget-allocation rule. With $C_{\max} = \frac{1}{2} N q_{\max}$, the holder can take roughly half the horizon at full rate, and a flat threshold takes exactly the dates on which the spot clears a common level. Since the spot is the seasonal curve plus quickly reverting noise, those dates concentrate at the winter peak, where $\bar{C}(t)$ indeed climbs fastest. Thus, seasonal timing is achieved by a flat threshold, not a seasonal one. Section 6.4 quantifies the resulting exercise calendar.

The mean path is one trajectory yet the policy is defined over all of (t, S, C). Figure 8 shows the policy at six fixed capacity levels (0, 525, 1050, 1120, 1190, 1250), each panel clipped to the feasible time window of its state, together with the full boundary surface. Two regimes organise the panels, split by $C_{\min}$. Below the floor, the obligation dominates. At $C = 0$ the threshold sits beneath the strike from shortly after the start, averaging

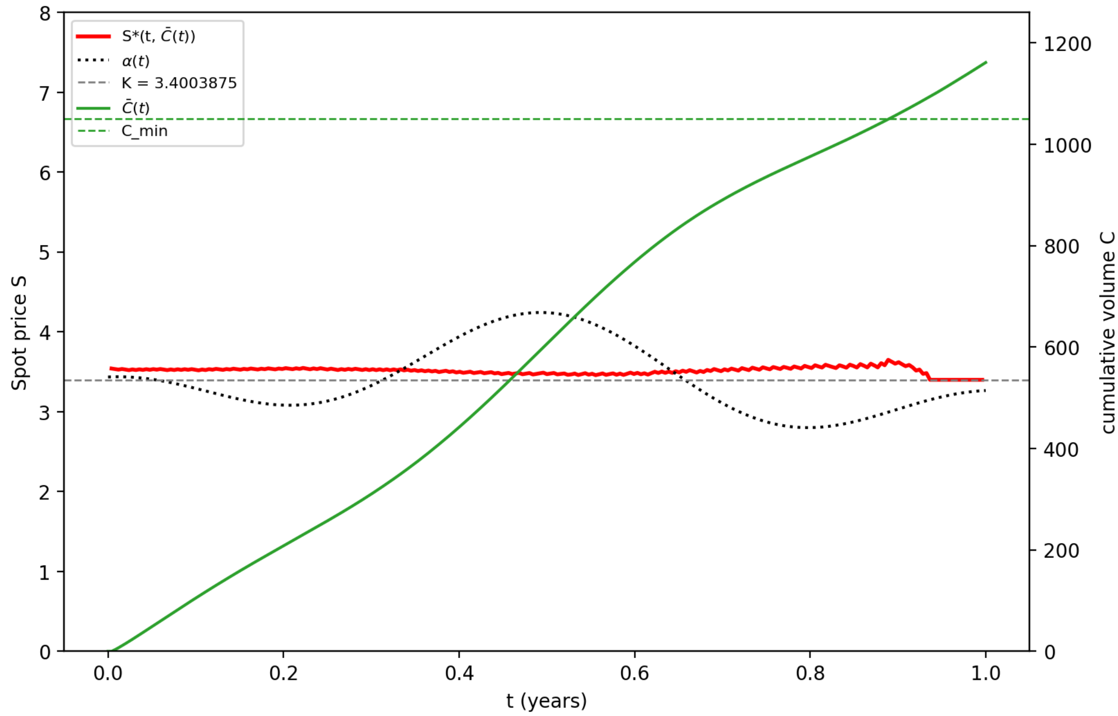


Figure 7: The mean exercise boundary, over 250,000 simulated paths.

\$2.66 across the panel and running about \$0.93 below the seasonal curve, so a holder who has taken nothing exercises at a loss rather than defer. Deferral only increases future obligation, and at $t = 0.587$ exercise of the full local cap is compelled at any price. Past that date a holder still at $C = 0$ could not reach the floor, so the state cannot arise under an admissible strategy. At $C = 525$ half the obligation has been served, and the effect of the receding floor is substantial. The threshold starts near \$4.2 at the panel's first feasible date rather than beneath the strike, and the forced onset shifts to $t \approx 0.79$. Above the floor, scarcity takes over. From $C = C_{\min}$ the panels run to expiry with no forced region, and the threshold rises with each panel, from about \$5.3 at the left edge of the $C = 1050$ panel to \$6.7 at $C = 1250$ where each unit already taken raises the price demanded for the next. Fixing an interior date isolates this effect. At $t = 0.5$ the threshold sits at about \$1.74 while the exercised volume is low, far under the strike, then rises convexly through it, reaching \$6.73 one unit short of exhaustion, a rise of roughly \$4.99 across the capacity axis. The fewer exercise rights remain, the higher the price the holder demands to part with one. Above the floor the threshold slides onto the strike as expiry nears, steeply so at high C, for the same reason the mean-path boundary does. The opportunity set shrinks and fast reversion leaves nothing to wait for. Below the floor the end behaviour is reversed. Here, the threshold falls further beneath the strike as the unmet obligation compels exercise.

The colour of the panels is close to binary, $q^* = 0$ or $q^* = q_{\max}$. Intermediate volumes are rare and appear as a narrow band along the boundary where the budget is tight. The surface plot assembles the slices into one object and shows the two regimes at a glance.

The trough beneath the strike across the obligation region, and the ridge climbing toward $C_{\max}$ where scarcity rules.

The surface reduces to a single rule the holder can use to act optimally. A holder should track the cumulative volume and exercise the full local cap whenever the spot clears the boundary. At the baseline that threshold lies within \$0.15 of the strike at the capacities a representative holder visits, widening only when the budget runs low or the obligation is far from met. This single-threshold, near-bang-bang structure is verified in Section 6.2 and traced against the model parameters in Section 6.3.

6.2 Structural Properties

Section 6.1 characterised the optimal policy and described when a holder should exercise. This section verifies that each feature seen there is a structural property of the optimal policy rather than noise. We examine five properties. Monotonicity of the action in the spot price, monotonicity of the boundary in cumulative volume and in time, the near bang-bang form of the optimal volume, and the forced-exercise region. Each diagnostic runs over the reachable state space at the baseline parameters of Table 1, on exercise days and excluding the terminal date, where no continuation exists.

Monotonicity in S

The exercise boundary $S^*(i, C)$ of Section 6.1 is well defined only if the policy splits the price axis cleanly. Specifically, a no-exercise region below a single threshold and an exercise region above it. This requires the optimal action to be monotone in the spot price, so that once exercise becomes optimal as S rises, it never reverts to waiting. Formally, for each reachable (i, C) the exercise set is an upper interval in the node index,

$$\big\{j : q^*(i, j, C) > 0\big\} = \big\{j : j \geq j^*(i, C)\big\} \tag{58}$$

for some threshold node $j^*(i, C)$, with $q^* = 0$ below it and $q^* > 0$ at or above it. Since $S_{i,j} = \alpha(t_i) + X_j$ increases with j, we have monotonicity in S, and the boundary is $S^*(i, C) = \alpha(t_i) + X_{j^*(i,C)}$.

This property is the one the boundary extraction of Section 4.6 presupposes. Taking the lowest exercising node as the boundary is meaningful only when every node above it exercises as well. We therefore verify (58) directly. For each reachable voluntary (i, C) we read the indicator $\mathbf{1}[q^*(i, j, C) > 0]$ across the reachable nodes ordered by increasing S and check that it is non-decreasing, changing from 0 to 1 at most once.

All 183,161 voluntary states checked satisfy (58) so the policy is single-threshold everywhere. The boundary of Section 6.1 is an exact description of the policy rather than an approximation, and the extraction rule is justified.

Monotonicity in C

The threshold rises with cumulative volume, so $S^*(i, C)$ is non-decreasing in C at every date. Section 6.1 identified two mechanisms behind this, the receding floor obligation below $C_{\min}$ and scarcity above it, and the surface of Figure 8 displays the combined effect climbing from the sub-strike trough at low C to the ridge at $C_{\max}$. At $t = 0.5$ the boundary does not clear the strike until $C \approx 580$, roughly the midpoint of the obligation region.

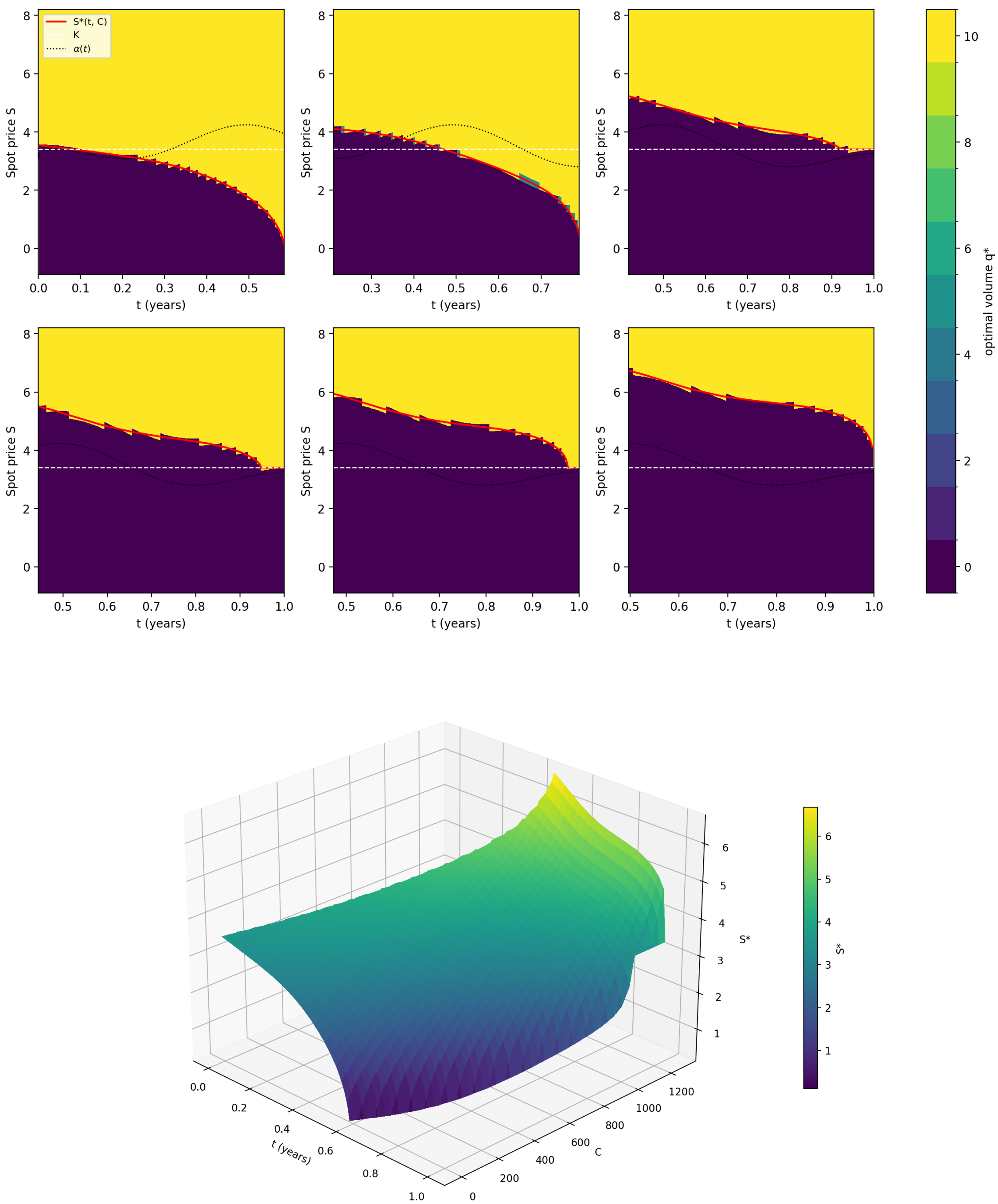


Figure 8: Top: optimal exercise volume q^* over (t, S) at six fixed cumulative volumes C (0, 525, 1050, 1120, 1190, 1250). Bottom: the boundary surface $S^*(t, C)$ over the full state space.

We verify the property at every exercise date. For each i we check that the interpolated boundary $S^*(i, C)$ is non-decreasing in C across the voluntary region, where the boundary is defined. The forced region is excluded since no voluntary threshold exists there. Monotonicity holds at every one of the 251 dates checked, up to a numerical tolerance. On the interpolated boundary the largest decrease anywhere in the surface is 0.0013 USD/MMBtu, approximately 0.9% of one price step $\Delta X = 0.139$. On the grid-snapped boundary every exception is exactly one price step, a symptom of node snapping, and they occur at 17 of the 251 dates. No violation is visible at the grid's granularity, so the rise is a property of the whole horizon rather than an artefact of any particular date.

Monotonicity in t

The threshold decreases as time increases, since at a fixed cumulative volume $S^*(i, C)$ is non-increasing in i across the voluntary region. The boundary a holder faces at any capacity level only descends with time. The reason is the same one as the flat mean-path boundary of Section 6.1. Waiting is priced by the chance of a higher spot to exercise into before the opportunity set closes, and both shrink with time. Fewer dates remain, and fast reversion pulls any deviation back before it can be taken advantage of, so the price demanded to exercise now rather than later tends toward the strike. This is the decline in each panel of Figure 8. It is worth separating from the sawtooth rise seen along the simulated paths of Figure 6, which is the state $C(t)$ jumping to higher capacity slices as volume is exercised instead of a fixed slice rising. Along a path the two effects compose, and the rise of the slice-switch dominates the decline within each slice.

For each C we take the interpolated boundary over its voluntary window, the consecutive exercise dates at which a threshold exists, from the first date to the last before the boundary terminates in the always-exercise and forced regions, and check that $S^*(i, C)$ is non-increasing in i up to the same tolerances as the check in C. Monotonicity holds on every one of the 1260 slices, with no slice exceeding the interpolation-noise tolerance and no gap in any voluntary window. The maximum increase anywhere in the surface is 0.0039 USD/MMBtu, under 3% of the price step ΔX, and three quarters of the slices are monotone with no increase at all.

The size of the decline depends on C. For $C = 0$ the opportunity window ends early, at $t \approx 0.58$ just before the forced onset, and over it the threshold collapses from \$3.54 to \$0.14, a fall of \$3.40. This is the steepest decline anywhere in the surface. The obligation is what drives it. As the forced frontier approaches, the price at which a holder who has taken nothing prefers to wait falls toward zero, since deferral buys nothing but a deeper forced state. Above the floor the windows run to expiry and the declines grow with scarcity, from \$2.23 at $C = 1050$ to \$3.04 at $C = 1250$, the slices whose thresholds start highest and have the furthest to fall. The two mechanisms are therefore distinct in kind. Below the floor the threshold is pushed down by an obligation that binds harder with time, and above it the threshold falls because the scarcity premium it carries has nowhere left to be spent.

Bang-bang structure

A control is bang-bang when it takes only its extreme values, here $q^* = 0$ or $q^* = q_{\max}$, and never an intermediate volume. The baseline policy is close to this form. The reason is the linearity of the immediate payoff. At (i, j, C) the one-step objective is

$$q\xi(S_{i,j}) + e^{-r\Delta t}\mathbb{E}_{\mathbb{Q}}\big[V_{i+1}(S', C+q) \mid (i,j)\big], \tag{59}$$

whose first term is linear in q and whose second depends on q only through $C + q$. Were the continuation value linear in cumulative volume, the whole objective would be linear in q and maximised at an endpoint of the feasible set $\mathcal{Q}_i(C)$, giving pure bang-bang. An interior volume is optimal only where the continuation value is concave enough in C for the declining marginal value of an extra unit to offset the constant immediate gain. That concavity is pronounced where the remaining budget is scarce, near $C_{\max}$ and near the forced floor, which is where the interior actions appear.

We quantify this over the reachable states. The optimal action is $q^* = 0$ at a fraction 0.489 of states, $q^* = q_{\max}$ at 0.505, and strictly interior at 0.006. Restricted to voluntary states the interior fraction is 0.005. The interior states concentrate in the later half of the contract, with median date $t = 0.70$, and cluster where the budget binds, predominantly towards $C_{\max}$ with a median cumulative volume of 1212 and an upper quartile of 1254. They form a band a few price nodes wide along the boundary, with roughly 62% lying within two nodes ($\approx$ \$0.28) of the threshold itself, visible in the $C = 1120$, $C = 1190$, and $C = 1250$ panels of Figure 8.

The forced-exercise region

The take-or-pay floor forces exercise near expiry. When too few dates remain for the local cap to carry the cumulative volume to $C_{\min}$, the holder must exercise regardless of price, in accordance with (23). In this region the holder exercises even below the strike, the prediction of Section 3.4.

We confirm the region directly. The forced set derived from the admissible actions matches the analytic condition (23) exactly, at every one of its 1050 onset states. Forced exercise first appears at $t \approx 0.587$ ($i = 148$) at $C = 0$, consistent with $(252 - 148)\, q_{\max} = 1040 < C_{\min} = 1050$, and the set extends over $i \in [148, 252]$, $C \in [0, 1049]$, widening as expiry approaches. The compulsion is anticipated well before it binds. The threshold at $C = 0$ slides from \$1.00 at $i = 140$ to \$0.14 at $i = 147$, the last voluntary date, so the holder is already exercising at almost every reachable price by the time the constraint takes the choice away. The onset is a limit the policy approaches rather than a discontinuity it meets. Within the forced region exercise occurs at spot prices as low as -\$4.58, realising the signed payoff of Section 3.4.

To conclude, monotonicity in S is the property that reduces the policy to the threshold rule of Section 6.1 without loss against the full policy. The other four describe where the threshold sits and how it moves, ordering the surface in both arguments. The boundary rises in C at every date and falls in t on every slice, so the ridge of Figure 8 descends toward expiry along every capacity level while climbing toward exhaustion at every date. Section 6.3 traces how these properties respond to the model parameters.

6.3 Parameter Sensitivity

We examine how the optimal policy moves when one parameter is varied and the rest are held fixed. For each sweep the dynamic program is re-solved at every value, with the remaining parameters at the baseline of Table 1, and the boundary is extracted for comparison. We compare fixed capacity slices rather than tracing the mean path $\bar{C}(t)$, since the mean path itself moves with the parameter, and a fixed slice isolates the response of the boundary surface from the response of the path through it. Section 6.2 showed that the surface is governed by two distinct forces, the obligation below $C_{\min}$ and scarcity above it, so we report one slice for each, $C = 0$ and $C = C_{\min}$. Where the combined effect is itself of interest, we additionally plot the boundary along each parameter value's own mean path. To compare across parameter values, each boundary is reduced to its mean level over the voluntary exercise dates of that slice, written $\bar{S}^*(C)$. This is a summary statistic of the policy which measures where the threshold sits, averaged over the year. The section first investigates the optionality margin parameters, then the take-or-pay obligation, and last the payoff level of Section 6.1.

Reversion speed κ

The reversion speed governs how fast a deviation from the seasonal level decays, and therefore how much waiting is worth. Figure 9 overlays the boundary along each parameter value's own mean exercised path, the object a representative holder faces, and Figure 10 shows the mean boundary level as a function of cumulative volume, one curve per κ.

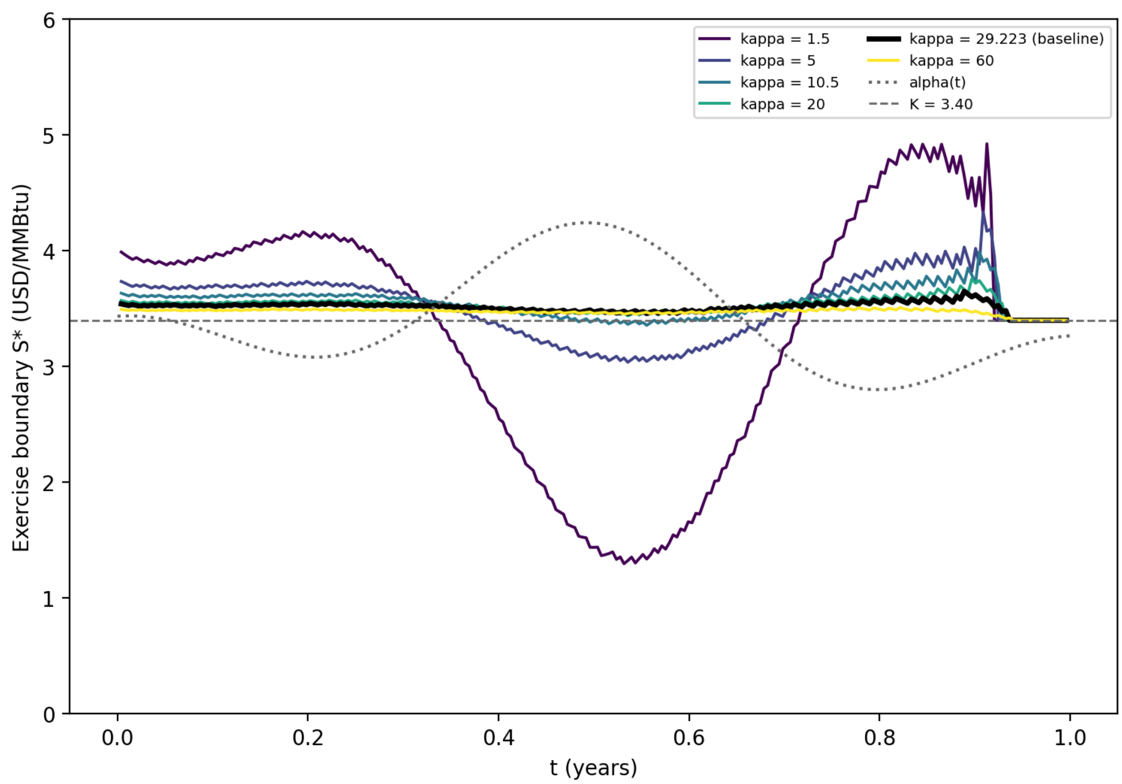


Figure 9: The exercise boundary along each κ's own mean exercised path $\bar{C}(t)$.

When reversion is slow, a deviation persists, so the price a holder sees today carries information about the price he will see later, and the boundary spreads far apart across

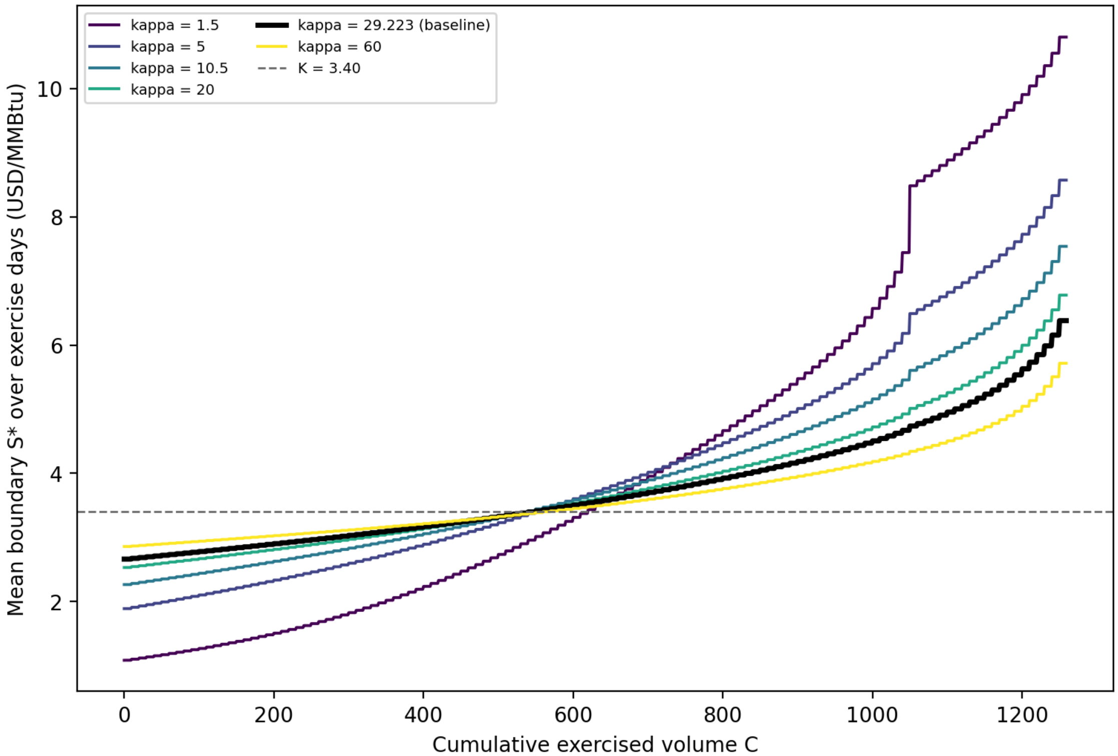


Figure 10: The mean boundary level $\bar{S}^*(C)$ against cumulative volume, one curve per κ.

capacity. Above the floor, persistence means a holder who waits keeps the chance of a far better price, so he exercises only well above the seasonal level, and at $\kappa = 1.5$ the mean boundary at $C = C_{\min}$ stands at \$8.48. Below the floor the same persistence works the other way. A holder who is behind on the obligation and sees a low price cannot expect it to recover before the deadline, so he takes volume at almost any price, and at $\kappa = 1.5$ the mean boundary at $C = 0$ falls to \$1.09, more than two dollars beneath the strike. As κ rises the deviation reverts within days, neither the promise of a better price nor the threat of a persistent bad one survives, and the two slices close on the strike from opposite sides, reaching \$2.66 and \$4.74 at the calibrated $\kappa = 29.2$ and \$2.86 and \$4.34 at $\kappa = 60$. Reversion speed therefore does not shift the boundary about the strike, it spreads it. The obligation slice and the scarcity slice separate as κ falls, and converge on the strike from opposite sides as it rises. Figure 10 shows this directly, the curves pivoting about $C \approx 550$ at a level close to the strike, so that a lower κ steepens the slope of $\bar{S}^*(C)$ in cumulative volume. At that pivot capacity neither force dominates and κ has nothing to act on.

Figure 9 shows what this spread means for a holder. At the calibrated speed the boundary along the mean path is flat, sitting a few cents above the strike for most of the year and locking onto it near $t \approx 0.93$, and at $\kappa = 60$ it is flatter still. At $\kappa = 1.5$ it is not flat at all. It swings by roughly \$3.5 over the year, from about \$1.4 near $t \approx 0.55$ to about \$4.9 near $t \approx 0.85$, and the swing runs in antiphase with $\alpha(t)$.

This is the seasonal timing of Section 6.1 made explicit in the policy. When a deviation persists, the holder cannot rely on a spike arriving at the date he wants it, so he books his volume against the part of the price he can predict, exercising almost unconditionally through the winter peak and refusing all but the largest deviations through the trough. The swing shrinks smoothly as κ rises, muted but visible at $\kappa = 5$ and nearly gone by $\kappa = 20$.

Reversion speed therefore decides where the seasonality of the asset is expressed. At slow reversion it is expressed in the boundary, which carries an inverted copy of the seasonal curve. At the calibrated speed it is expressed only in the exercise calendar. The boundary is flat and the spot simply clears it more often in winter, which is the mechanism Section 6.1 identified and Section 6.4 quantifies. κ therefore acts on the form of the policy, not its height. It decides whether seasonality appears in the boundary or only in the calendar, a qualitative switch rather than a shift in level.

The reversion speed is the least robust calibrated input, so its effect on the policy is what decides whether that fragility matters. Excluding the February 2021 and January 2026 spikes roughly halves the estimate, to $\kappa \approx 10.5$. The response is convex in κ and most of its range is spent below $\kappa \approx 10$, so the spike exclusion does not move the policy across regimes. Both estimates sit in the fast-reverting regime with a boundary that is nearly flat and near the strike, and neither recovers the seasonal shape of Figure 9. The shift within the regime is not negligible. At $\kappa = 10.5$ the mean boundary is \$2.27 at $C = 0$ and \$5.61 at $C = C_{\min}$, against \$2.66 and \$4.74 at the baseline, so the estimate's uncertainty moves the threshold by \$0.40 and \$0.87, roughly a quarter of the \$1.58 and \$3.75 swings from $\kappa = 1.5$ to the baseline. The regime conclusion is robust to the spike exclusion. The position of the boundary within the regime is not.

Volatility σ

Volatility scales the size of the deviations, and therefore the value of the optionality the holder exercises. It spreads the boundary about the strike in the same way as κ, but with the sign reversed, since a larger σ widens the distribution of future deviations and makes both a high price worth waiting for and a low price worth fearing more. Above the floor the mean boundary rises with σ, from \$3.83 at $\sigma = 2.5$ through \$4.74 at the calibrated $\sigma = 12.75$ to \$6.01 at $\sigma = 25$. Below it the mean boundary falls, from \$3.17 through \$2.66 to \$2.01. Both arms are near-linear in σ and neither flattens over the range examined. The calibrated value sits on a straight line rather than on a tail.

The contrast with κ at the calibrated point is one of curvature, not of direction. Both parameters spread the boundary about the strike and locally σ does so about twice as strongly. A ten percent move in σ shifts the mean boundary by $-\$0.067$ at $C = 0$ and $+\$0.131$ at $C = C_{\min}$, against $+\$0.032$ and $-\$0.066$ for κ, the signs opposed because faster reversion and larger deviations spread the boundary in opposite directions. The difference lies away from the calibrated point. The slope of the κ response decays as κ grows, so an error on the high side costs little. The σ response is linear throughout, so an error in σ transmits in proportion to its size wherever it falls. Volatility is the estimate whose error reaches the exercise strategy undamped.

Take-or-pay floor $C_{\min}$

The floor sets the obligation, so it moves the forced region and the boundary below $C_{\min}$ rather than the threshold above it. A higher floor compels exercise sooner. The forced-onset time, the earliest date at which exercise becomes compulsory at $C = 0$, falls linearly as $C_{\min}$ rises, from $t = 0.794$ at $C_{\min} = 525$ to $t = 0.528$ at $C_{\min} = 1200$, through the baseline $t = 0.587$ at $C_{\min} = 1050$, matching the forced condition (23) at every value swept. At $C = 0$ the last voluntary index is $N - C_{\min}/q_{\max}$ and the onset is the date after it, so the onset time falls by $1/(Nq_{\max})$ per unit of floor.

The sweep also isolates the obligation as the cause of the sub-strike threshold of Section 6.1. The mean boundary at $C = 0$ falls monotonically with the floor, from \$3.45 at $C_{\min} = 0$ to \$2.57 at $C_{\min} = 1200$, a spread of \$0.88, and the value at $C_{\min} = 0$ sits five cents above the strike. Remove the obligation and the $C = 0$ threshold returns to the strike. Everything the baseline $C = 0$ slice does beneath K is therefore the floor's doing, and none of it is optionality. Above the floor the mean boundary is untouched, identical at every $C_{\min}$ from 0 to 1050, since the value function at $C \geq C_{\min}$ cannot see a constraint that has already been satisfied. It moves only at $C_{\min} = 1200$, where $C = 1050$ now falls below the floor, so the slice acquires a forced region and its mean is no longer taken over the same voluntary dates as at the lower floors. The floor governs the obligation and leaves the optionality margin alone, the opposite of κ and σ, which spread the margin and leave the forced onset exactly where it is.

Strike K

The strike enters the payoff directly, so moving it shifts the level against which the holder measures a favourable price. A plain option's exercise threshold would follow it almost one for one. This contract's does not. Restricted to the voluntary dates common to the strikes examined, the mean boundary shifts by 0.11 per unit of strike at $C = 0$ and by 0.35 at $C = C_{\min}$.

The attenuation is the take-or-pay structure. A right that will be exercised regardless of price is a forward, not an option, and a forward's timing decision is invariant to the strike, since K enters every date's payoff equally and cancels from the comparison. Only the part of a state's value that is genuinely optional responds to K, so the shift measures how much optionality the state carries. It is near zero at $C = 0$, where the floor will take the volume whatever the price. It is larger at $C = C_{\min}$, where the rights are unrestricted, and larger still on the out-of-the-money arm, where more of the continuation value lies in the possibility of not exercising at all. The strike therefore does not shift the boundary consistently. It shifts it in proportion to the optionality that survives the obligation.

To conclude, the parameters separate cleanly by the force each moves. Reversion speed and volatility spread the boundary about the strike, driving the obligation slice down and the scarcity slice up. The take-or-pay floor moves the obligation and the forced region alone, and the strike shifts the boundary in proportion to the optionality the obligation leaves intact. The daily cap $q_{\max}$ is not an independent force. It enters only through $C_{\min}/q_{\max}$, which sets the forced onset, and $C_{\max}/(Nq_{\max})$, which sets the budget, so it acts on both the obligation and the scarcity margin at once and is reported as a check rather than swept.

The two margin parameters differ away from the calibrated point rather than at it, but κ matters most through the regime it identifies. Fast reversion makes the boundary flat, near the strike, and blind to the season, delegating the seasonal timing entirely to the exercise calendar, which Section 6.4 continues on.

6.4 Seasonal Behaviour

The preceding sections characterised the policy over the entire state space. This one investigates where exercise concentrates on the calendar. The two findings appear to contradict. The threshold of Section 6.1 ignores the seasonal curve, yet exercise concentrates in winter where gas is more valuable. Figure 11 resolves it. It bins the volume exercised across 250,000 simulated paths under the optimal policy into 63 four-day bins, split into a voluntary and a forced component, with the seasonal level overlaid. An exercise event is counted as forced when the remaining budget has no slack, $C + (N - i)\, q_{\max} \leq C_{\min}$, so that any admissible action fills the local cap and no discretion remains. This includes the knife-edge equality, where the policy defers as long as feasibility allows and rides the constraint to expiry, though most forced volume falls strictly inside the region of Section 6.2, on paths whose cumulative volume fell far enough behind that the floor was already unreachable with slack to spare.

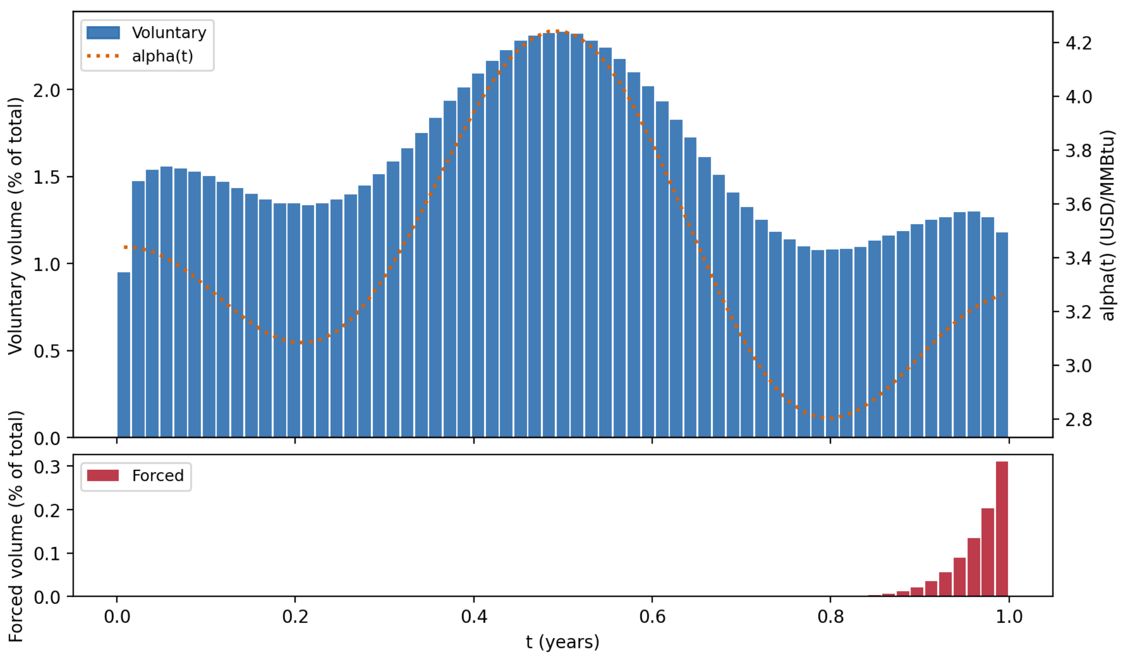


Figure 11: Exercised volume by date as a share of the total under the optimal policy, split into voluntary (top) and forced (bottom) components, with the seasonal level $\alpha(t)$ overlaid on the voluntary panel.

Voluntary exercise tracks the season without a seasonal threshold. Per-date voluntary volume moves with $\alpha(t)$ with a correlation of 0.95, peaking where the seasonal curve does and thinning in the troughs. The high-α window $t \in [0.32, 0.65]$, a third of the year, takes 44% of the voluntary volume, exercised at a mean moneyness of +\$1.40 and a volume-weighted mean date of $t = 0.48$. The mechanism is the one Section 6.1 already identified, where the spot is the seasonal curve plus fast-reverting noise, so a flat boundary is cleared most often when the curve itself is high. The tilt is moderate rather than extreme because the contract leaves little room for selectivity. The floor alone mandates $C_{\min}/q_{\max} = 105$ full-rate days, about 42% of the horizon. With so many dates needed regardless, seasonal timing can only tilt the allocation toward winter and not concentrate it there.

The forced component is the calendar footprint of the take-or-pay floor, and it is small. It carries 0.9% of total volume, clusters hard against the deadline, with 95% falling in the final tenth of the year and a volume-weighted mean date of $t = 0.96$, and is exercised at a mean moneyness of $-\$1.16$. These are prices below the strike that no voluntary rule would accept, the below-strike exercise of Section 6.2 realised in equilibrium, on the minority of paths whose winters were too cheap to lift the cumulative volume clear of the floor. The losses are comparable in size to the typical voluntary gain, about \$1.16 against \$1.40, but they fall on so little volume that they are a tail of the distribution rather than a feature of the typical contract.

The two components separate cleanly in the figure, seasonal voluntary exercise priced above the strike and a deadline-driven forced tail priced invariant to the strike. The difference between them is the section's finding. The floor mandates 42% of the horizon at full rate and, as Section 6.3 showed, dominates the boundary below $C_{\min}$. However, under 1% of volume is ever taken under compulsion. The floor shapes the policy by anticipation. The threshold slides beneath the strike early enough that a holder meets the obligation voluntarily on all but the cheapest paths, the same slide toward the forced onset that Section 6.2 observed, seen here in aggregate. Seasonal timing, in this regime, is therefore an output of the model rather than an input. The calibrated reversion speed makes the threshold flat, the seasonal curve pushes the spot across it, and the floor lowers the threshold in advance far more often than it forces the holder's hand.

6.5 Comparison to Naive Policies

The optimal policy is a surface over three state variables. This section measures what that structure is worth against simpler strategies a holder may consider instead. Each benchmark removes one component of the optimal policy's intelligence, so the gap it leaves quantifies the value of the component removed. All policies are evaluated by Monte Carlo on the same 250,000 simulated paths under common random numbers, so the gaps are paired and their standard errors are far smaller than the standard errors of the values themselves. The harness is validated against the lattice in Section 6.6. Every benchmark is made feasible by construction.

We evaluate four strategies. The pro-rata policy spreads the budget evenly, taking $\frac{C_{\max}}{N} = 5$ units every day regardless of price, and so carries the contract structure with no price intelligence at all. The deterministic seasonal policy exercises at the local cap on the $\frac{C_{\max}}{q_{\max}} = 126$ dates on which the seasonal curve $\alpha(t)$ is highest, fixed in advance, and so responds to the predictable component of the price while ignoring the realised deviation. The fixed threshold at K policy exercises the local cap whenever the spot clears the strike, the myopic rule that takes any immediately profitable unit. The best constant threshold generalises it, exercising above a level $\bar{S}$ fitted by grid search on an independent set of 250,000 paths, and so is the strongest policy available in the class the mean-path boundary of Section 6.1 visually resembles. What none of the four possesses is the surface itself, the threshold's descent in t, its rise in C, and its anticipation of the obligation.

Table 3 orders the rules by the intelligence they carry, and the ladder is steep at the bottom and shallow at the top. The pro-rata value is statistically indistinguishable from zero. The strike sits almost exactly at the level of the average seasonal curve, so a rule

Table 3: Naive policies against the optimal policy on 250,000 common paths.

Policy	Value	Gap (%)	Forced (%)
Pro-rata	−7.7	100.5	0.0
Deterministic seasonal	411.2	73.9	0.0
Fixed threshold at K	1499.5	4.9	2.8
Best constant threshold	1505.0	4.5	4.0
Optimal (DP)	1576.5	0.0	0.0

that buys unconditionally collects the signed payoff on both sides of K in nearly equal measure, and the entire value of the contract lies in choosing dates rather than in holding volume. The deterministic seasonal rule recovers about a quarter of the optimal value. Concentrating the budget on the winter peak captures the spread between the high season and the strike, which prices the predictable component of the exercise calendar of Section 6.4, and its shortfall shows that most of the value lies in responding to the deviation rather than to the season.

The jump to the threshold rules is the section's finding. A fixed threshold at the strike already recovers 95.1% of the optimal value, and the best constant threshold 95.5%. A single number, held fixed for the whole year at every capacity level, captures all but \$71 of a \$1576 contract. This confirms the flat mean-path boundary of Section 6.1 from an entirely independent direction. In the fast-reversion regime the optimal threshold barely moves along the path a typical holder follows, so a rule that does not move it at all loses little. The fitted level makes the corroboration exact. The grid search, run on separate paths with no access to the dynamic program, selects $\bar{S} = 3.49$, nine cents above the strike, against the mean-path boundary level of \$3.51 found by the lattice. Two unrelated computations place the threshold at the same level.

The residual 4.5% is then the value of the surface over its best constant approximation, and the forced exercise column shows where it lives. The constant-threshold rules meet the floor by compulsion on 3 to 4% of their volume. The obligation-driven descent of the boundary below $C_{\min}$ is therefore not lost to them but imposed on them, crudely and late, whereas the optimal policy anticipates it and buys ahead of the deadline at better prices. What the constant rules genuinely lack is the scarcity rise near $C_{\max}$ and the endgame slide onto the strike, the refinements of Sections 6.2 and 6.3. The gap is statistically sharp, at more than two hundred paired standard errors, but economically modest, and that modesty is itself the result. The regime that makes the optimal boundary flat is the same regime that makes flatness cheap to impose.

6.6 Validation

The results of this chapter rest on a chain of three objects. The value function is solved by backward induction, the policy is its argmax at every state, and the boundary is extracted from the policy by interpolating the zero of the exercise advantage. Each link can fail independently, and this chapter's own history supplies the example. A sign error in the exercise payoff produced a value function that overstated the price by 3.55%, while every internal consistency check of the solver passed. This section therefore validates each link separately, in order. The value function is checked against closed forms, analytic bounds, and its own convergence. The policy is checked by realising its value in simulation, by perturbing it in directions an implementation error would move it, and against an independent optimiser. The extracted boundary is checked for exactness of the extraction rule and health of the interpolation.

One bounding argument underlies several of the checks and is stated once here. Any admissible policy, evaluated without lookahead by simulation, realises a value that is a lower bound on the true optimum, since the optimum could always have followed that policy. The lattice price should therefore sit at or above those realised values, and within Monte Carlo noise of the values realised by its own policy. An independent optimiser constructs a second lower bound with no access to the lattice. Two lower bounds from unrelated methods that agree pin the optimum from below far more convincingly than either alone, since an error in either would surface as disagreement. All simulation checks share one evaluation procedure and one set of 250,000 paths under common random numbers, the same paths as the benchmarks of Section 6.5, so every comparison between policies is paired and its standard error reflects only the difference between them.

The value function

The sharpest test of the pricing methodology involves no optimisation at all. Setting $C_{\min} = C_{\max} = (N{+}1)\, q_{\max}$ forces the full local cap at every date, no choice remains, and the price is a closed form under the arithmetic OU dynamics,

$$V = \sum_{i=0}^{N} e^{-rt_i}\, q_{\max}\big(\alpha(t_i) + X_0 e^{-\kappa t_i} - K\big), \tag{60}$$

which exercises the payoff sign, the discounting, the seasonal curve, and the OU drift in isolation, and is derived in Appendix D.1. Two conditions make the comparison exact. First, $Nq_{\max}$ leaves one date of slack under the $N{+}1$-date convention, so genuine forcing requires $(N{+}1)q_{\max}$, the setting used above. Second, the lattice snaps X_0 to the nearest node before solving, so the dynamic program prices a contract that starts at the snapped deviation. The closed form must be evaluated at that same snapped point, which separates discretisation error from a mismatch in the initial condition. Evaluated this way, the solver agrees with the closed form to within 0.02% at every resolution level. The same configuration also prices the snap itself. Every date exercises, so an error in the starting deviation compounds over the whole horizon, and the gap between the closed form at the true and at the snapped X_0 reaches \$20.59 at the three coarsest resolution levels, where the price step exceeds $|X_0|$ and the snap lands at zero, and \$4.78 at the production resolution. This is the identified source of the noise in the price ladder below. This configuration is also the one on which the historical payoff error fails, a fully forced contract with dates below the strike. That error has an exact economic interpretation. Clipping the voluntary payoff at zero let the solver satisfy the floor at no perceived cost, pricing the contract as if the obligation were absent, so the defective price equals the correct $C_{\min} = 0$ price. The \$55.84 it overstated the $K_{\text{sub}} = 1$ price by is exactly the cost of the take-or-pay floor.

Two degenerate contracts check the optimisation itself against analytic references. Decoupling the constraints entirely, $C_{\min} = 0$ and $C_{\max} = (N{+}1)q_{\max}$, makes every right independent, and the swing price must equal a strip of independent European values date by date. We find it does, to within 0.04% at every resolution level above the coarsest and 0.33% at $K_{\text{sub}} = 1$. Because the rights are independent here, the correct policy is simply to exercise each date whenever its own advantage is positive. Comparing the solver's decisions against that rule state by state, disagreements appear only at capacity levels unreachable from the initial state, always in the direction of the solver exercising less than the rule at a marginal advantage, and never at a reachable state. Restoring the real ceiling $C_{\max} = 1260$ while leaving $C_{\min} = 0$ couples the rights through the budget alone, and the price is then bracketed analytically. The unconstrained strip bounds it above, since relaxing the ceiling only adds value, and the strip restricted to the $\frac{C_{\max}}{q_{\max}} = 126$ highest-value dates, committed in advance, bounds it below, since pre-commitment is an admissible policy. The solved price of this floor-free contract sits inside this sandwich at every resolution level, at the production resolution between 1059.52 and 1673.59 at a value of 1613.05. Reducing instead to a single right, $C_{\max} = q_{\max}$ and $C_{\min} = 0$, gives a Bermudan option on one lot, bounded below by the best single European and above by the sum of all of them. Both bounds derive from the same OU law the lattice discretises, so this reduction is a consistency bracket rather than an independent cross-solver check, and its upper arm is loose. The independent optimiser of the following subsection, run on this same configuration, tightens the lower bound from \$1.15 to within \$0.002 of the solver. Against a lattice value of 3.327, the regression policy realises 3.325 at degree three and 3.326 at degree four, closing over 99.9% of the analytic bracket from below with no access to the lattice. The decoupled configuration finally supplies an exact reference for the strike sensitivity of Section 6.3. With no budget trade-off the boundary must sit exactly at the strike at every date and shift one-for-one with it, and the solver reproduces $S^* = K$ identically and $dS^*/dK = 1.0000$ on both arms at every resolution level. The damped slope of 0.35 at the baseline therefore measures the obligation, not the code.

Convergence in the lattice resolution closes the value-function checks. The price sequence separates into a converging scheme and a moving initial condition. On the three finest resolution levels the fitted convergence order is $p = 2.14$, with a Richardson-extrapolated limit of 1574.11. The production resolution level prices at 1573.74, \$0.37 below that limit, or 0.023%, the coarsest level is off by 0.07% and the finest by 0.011%. Across the full ladder the raw sequence is non-monotone, spanning 1571.7 to 1583.9, and the fully forced closed form above identifies the cause as the snap of the initial deviation, which moves between resolution levels as ΔX changes and is worth up to \$20.6 at the coarsest levels. From $K_{\text{sub}} = 32$ upward every level prices within 0.25% of every other. The boundary is tighter still, though it supports no order fit, since its successive differences are non-monotone at magnitudes below a tenth of a cent. The mean boundary level at $C = 0$ rises from 2.6404 at $K_{\text{sub}} = 1$, an error of 0.87%, and settles at 2.6635 from $K_{\text{sub}} = 64$ onward, every level from there agreeing to within 0.0002 USD/MMBtu and the production level matching the finest to a thousandth of a cent. The discretisation uncertainty in the price remains an order of magnitude larger than that in the boundary, and the boundary is the object the chapter's claims rest on.

The optimal policy

The checks of this subsection execute policies at the validation resolution $K_{\text{sub}} = 32$, at which the mean boundary level lies within 0.0003 USD/MMBtu of the production surface and the price within 0.08%, so their conclusions carry over to the production resolution. The first check realises the claimed value. The lattice's own policy is followed on 250,000 simulated paths and evaluated under the signed payoff, with the spot mapped to its nearest node for the policy lookup, a mean absolute snap of \$0.03. The realised mean is 1576.45 with a 95% confidence interval of [1574.32, 1578.57], sitting \$2.71 above the production lattice value of 1573.74: the policy realises more than the lattice claims, not less. This is the sharpest single check available, because a value function can overstate in ways no internal consistency test detects, and the only symptom is a price its own policy cannot realise. The historical payoff error had exactly this signature. The defective lattice claimed 1622.92 while its policy realised only 1525 to 1543 once the signed payoff collected the losses the solver believed free, and every internal check passed throughout.

The second check perturbs the boundary deliberately, and it is the only check that speaks to the chapter's central object rather than to the price. Every other diagnostic would pass under a boundary that was systematically misplaced by an amount too small to move the value, which is a live concern precisely because Section 6.5 showed that flattening the entire surface to a constant costs only 4.5%. The test replaces the policy by the threshold rule at $S^*(i, C) + \Delta(i, C)$ and evaluates each member on the same paths, so every gap is paired. Four families of deliberate error are examined, a level shift of one to eight price steps in both directions, tilts in C and in t pivoting at mid-range, and a shrinkage $\lambda S^* + (1 - \lambda)\bar{S}$ toward the fitted constant of Section 6.5 over $\lambda \in [0, 1.25]$. At a true optimum the envelope theorem makes the loss second order in the displacement, so the signature of health is a parabola with its maximum at the unperturbed policy, and this is what every family shows in Table 4. No member beats the dynamic program, the level family is maximised at zero displacement with a near-symmetric loss of \$15 to \$17 at one price step against paired standard errors of \$0.12, and the smallest tilts in either state variable cost \$5 to \$6 at paired standard errors near \$0.05, more than a hundred standard errors from zero. The shrinkage family is maximised at $\lambda = 1$, its over-steepened member $\lambda = 1.25$ losing \$2.74, so the optimum is interior in both directions, and its $\lambda = 0$ endpoint reproduces the best-constant value of Section 6.5 within Monte Carlo noise, connecting the two experiments continuously. Reconstructing the policy as a pure threshold rule costs $\$2.52 \pm 0.03$ against the full policy, about 0.16% of the value, consistent with discarding the interior volumes at the 0.5% of states that carry them together with the node snap of the lookup. The test is honest about its scope. It certifies local optimality in the tested directions and an unbiased extraction, the failure shapes an implementation error produces, and cannot exclude a superior boundary outside the spanned families. That exclusion rests on the construction of the dynamic program, which maximises over every admissible action at every state, together with the value-function checks above, and on the shape-free comparison that follows.

The third check is an independent optimiser. A least-squares Monte Carlo solver is built on the state (t, X, n) with n the remaining rights, exploiting the bang-bang structure verified in Section 6.2, with the take-or-pay floor enforced through the same admissibility rule as the lattice, regressions of realised future cashflows on a standardised polynomial basis at every date and rights level, coefficients fitted on an independent path set, and the induced policy evaluated out of sample on the common evaluation paths. Its realised value and the lattice policy's realised value are both honest lower bounds on the true optimum, obtained by unrelated methods, and they agree in an instructive way. Conditioning on

Table 4: Perturbation test. Paired loss against the unperturbed threshold policy on 250,000 common paths, in USD, with paired standard errors. Displacements in price steps ΔX at the validation resolution $K_{\text{sub}} = 32$.

Member	Loss	SE	Chg. (%)
Level ± 1	15.09 / 16.69	0.12	3.9
Level ± 2	59.78 / 64.22	0.23	7.5
Level ± 4	216.7 / 236.3	0.41	15.3
Level ± 8	637.7 / 736.3	0.65	30.5
Tilt C, ± 1	5.71 / 5.97	0.05	3.1
Tilt t, ± 1	5.07 / 5.60	0.04	3.0
Shrink λ=0.75	2.65	0.06	1.6
Shrink λ=0	69.78	0.25	7.9
Shrink λ=1.25	2.74	0.04	1.4

the continuous price, the regression policy realises \$1.27 more than the node-snapped execution of the lattice policy, a gap of 0.08% that is stable across basis degrees two to four. Handicapping the regression policy to the same snapped price information reverses the sign, leaving the dynamic program ahead by \$0.007 to \$0.018, at most two and a half paired standard errors from zero, at every degree. The excess is therefore the information cost of executing the policy from a discrete grid, not suboptimality of the policy itself, and under equal information neither optimiser materially outperforms the other, closing the bracket of the bounding argument from both sides. A dual upper bound in the manner of Andersen and Broadie, and an independent finite-difference solve of the HJB equation, would certify the remaining distance to the optimum and are noted as further work.

The extracted boundary

The extraction rule is exact rather than approximate. Section 6.2 verified the upper-interval property at all reachable voluntary states, so the exercise set is a half-line in the price at every state, the lowest exercising node is the boundary by construction, and the sub-node interpolation of the advantage refines the threshold strictly inside the bracketing interval the grid provides.

The interpolation reports its own diagnostic, the fraction of states at which the sub-node root-find fails and falls back to the grid-snapped value. Across every configuration solved in this chapter, the baseline and all 24 parameter-sweep runs, this fallback fraction never exceeds 0.36%, reached at the slowest reversion speed, and sits at 0.02% at the baseline resolution and 0.04% at the production resolution. A positive control establishes sensitivity. Handing the interpolator the correct solved arrays but a deliberately mismatched strike drives the fallback fraction to 100.0% and fires a warning. A low reading is therefore informative, since the check responds unambiguously when the boundary is genuinely misplaced.

The extracted level is finally corroborated from outside the lattice. The grid search of Section 6.5, a Monte Carlo computation with no access to the dynamic program, fits its best constant threshold at \$3.49, while the lattice extraction, with no simulation involved, places the mean-path boundary at \$3.51. Two unrelated computations locate the same level to within \$0.025, about 0.7%.

7 Conclusion

This thesis asked what the optimal strategy is for exercising a swing option on a seasonally mean-reverting asset. To answer it, we modelled the Henry Hub spot as an arithmetic Hull-White process, a deterministic seasonal level plus an Ornstein-Uhlenbeck deviation, calibrated the dynamics to ten years of spot history and the risk-neutral level to the forward strip, and solved the resulting stochastic control problem by backward dynamic programming on a sub-stepped trinomial lattice over time, price, and cumulative exercised volume. The output is the exact optimal policy at every discretised state, from which the exercise boundary is extracted at sub-node resolution.

The policy admits a compact description. At every reachable voluntary state the optimal action is monotone in the spot, so the policy reduces without loss to a single threshold surface $S^*(i, C)$, the lowest price at which the holder exercises. The surface is ordered in both arguments, rising in cumulative volume as scarcity and the receding obligation raise the price demanded for each remaining right, and falling in time as the opportunity set closes. The optimal volume is near bang-bang, with intermediate actions confined to a narrow band along the boundary where the remaining budget no longer accommodates a full local cap, near $C_{\max}$ and at the forced floor. The take-or-pay floor bends the threshold below the strike well before forced exercise starts, so the obligation is met by anticipation on all but the cheapest paths, and exercise at a loss is realised on only a small fraction of the volume.

The parameters separate cleanly by the force each moves. The reversion speed acts on the form of the policy rather than its height, deciding whether the boundary carries the season or delegates it to the exercise calendar. At the calibrated speed a deviation decays within days, timing optionality is nearly worthless, and the threshold a representative holder faces is flat and within cents of the strike, whereas at slow reversion the boundary carries an inverted copy of the seasonal curve. Because the response flattens at high reversion speed, the near halving of the reversion estimate under spike exclusion moves the boundary within the regime but not across it. Volatility transmits linearly and undamped, making it the estimate whose error reaches the policy in proportion. The floor governs the obligation and the forced region alone, and the strike shifts the boundary only in proportion to the optionality the obligation leaves intact.

Seasonality enters the exercise calendar without entering the threshold. Exercise concentrates in winter because the spot is the seasonal curve plus fast-reverting noise, so a flat boundary is cleared most often where the curve is high. Seasonal timing is therefore an output of the calibrated regime rather than a feature built into the rule.

Each link of the computation was validated separately. The value function agrees with closed forms and analytic brackets in degenerate configurations, the policy realises its claimed value in simulation and matches an independent least-squares Monte Carlo optimiser once both act on the same information, deliberate perturbations of the boundary lose value in every direction tested, and a grid-searched constant threshold locates the same level from outside the lattice. A constant threshold recovers nearly all of the contract value, so the regime that makes the optimal boundary flat also makes it cheap to approximate, and the remaining premium of the full surface lies in the scarcity rise near exhaustion and the anticipation of the obligation.

The model's limitations, among them the Gaussian tail, the single stochastic factor, and the deterministic seasonal level, are taken up together with extensions in Chapter 8.

8 Discussion & Further Work

This chapter collects the limitations of the thesis and the work they motivate. Each limitation is stated together with the cost it imposes on the conclusions and the extension that would remove it.

8.1 The Model

The spot model is a single Gaussian factor about a deterministic seasonal level, with constant reversion speed and volatility. Each of these ingredients is contradicted by evidence the thesis itself produced. The Ljung-Box rejection of residual whiteness in Section 5.5 points to a second, faster-reverting factor that the one-factor specification folds into the noise. The spike-exclusion rerun of the same section roughly halves both estimates, from $\hat{\kappa} = 29.2$ to 10.5 and $\hat{\sigma} = 12.75$ to 6.46, so the constant parameters are absorbing two storm episodes rather than describing a homogeneous regime. The Gaussian tail admits negative prices with probability peaking at 4.6% at the seasonal trough, and the deterministic seasonal level is itself an approximation that gas-specific evidence rejects in favour of stochastic seasonality [23, 30, 31].

The cost falls unevenly across the conclusions, and the sensitivity analysis of Section 6.3 locates where. The response of the boundary to the reversion speed flattens at high κ, so the regime conclusion, a flat threshold near the strike with seasonal timing delegated to the exercise calendar, survives even the halving of the estimate. The response to volatility is linear, so any misspecification of σ, including the share of it induced by treating spike variance as diffusive, reaches the policy in proportion. The negative-price mass distorts precisely the low-price region in which the forced-exercise losses of Section 6.4 are realised, so the magnitude of those losses is the least trustworthy quantity in the results.

The remedies are standard extensions, each purchasing realism at a structural price. A geometric specification restores positivity but destroys the additive threshold decomposition $S^* = \alpha(t) + X_{j^*}$ on which the entire analysis of Chapter 6 is built, a trade-off argued in Chapter 2. A jump-diffusion extension of the arithmetic form [25] separates the spikes from the diffusion, directly addressing the spike contamination of the calibration, and a second factor addresses the residual autocorrelation. We conjecture that the central finding is robust to these extensions, since any specification that preserves fast reversion of the diffusive component should reproduce a flat voluntary threshold, but this is a conjecture rather than a result, and verifying it under a jump-diffusion is the natural first extension.

8.2 The Calibration

The measure change of Section 2.5 is internally inconsistent by construction. The constant market price of risk is retained for the dynamics, where it preserves κ under $\mathbb{Q}$, and abandoned for the level, which the forward curve's stronger winter seasonality forces to be fitted separately. The cost is that the seasonal risk premium visible in Figure 1 is left unmodelled, and the preservation of κ rests on an assumption the level data contradicts. A deterministic periodic market price of risk $\lambda(t)$, calibrated jointly to the strip and the history, would restore consistency and let the model account for the seasonal premium it currently leaves unmodelled.

The window-stability analysis of Section 5.5 shows the rolling estimates ranging from 1.4 to 84.7 in $\hat{\kappa}$ and 2.2 to 27.8 in $\hat{\sigma}$, so the ten-year window is a choice rather than an estimate of a stable quantity, and the calibrated point is one draw from a wide range. The flattening response damps this for κ but the linear response transmits it for σ, which is therefore the estimate whose instability matters most. The reported standard errors are nominal and uncorrected for the residual autocorrelation, so they understate the uncertainty in $\hat{\kappa}$. Autocorrelation-robust or bootstrap errors would repair this. The constant interest rate is of little consequence at a one-year horizon.

8.3 The Numerical Method

Discretisation error is a measured quantity rather than an open one. The convergence ladder of Section 6.6 resolves the boundary, the object the conclusions rest on, to well below a cent at the production resolution, while the price carries a discretisation band an order of magnitude wider, traced to the snapping of the initial deviation to the grid. What the validation chapter does not supply is an upper bound. Every check approaches the optimum from below, so the true value is bounded below but not bracketed. Two extensions would close this and are left as further work. A dual upper bound in the manner of Andersen and Broadie would certify the remaining gap, and an independent finite-difference solve of the Hamilton-Jacobi-Bellman variational inequality would cross-check the entire lattice construction with no numerical structure in common.

8.4 Scope of the Findings

The structural properties of Chapter 6 are verified exhaustively over the discretised state space, but at one calibration and one contract. They are empirical regularities, not theorems, and the thesis makes no claim beyond the parameter point examined. Two programmes would extend the scope. The theoretical one is proof, extending the monotonicity and threshold results established for related formulations [10, 11, 17] and the bang-bang conditions of [7] to the variable-volume take-or-pay contract, converting the verified regularities of Section 6.2 into properties of the model. The empirical one is the mirror image of this thesis, holding the calibrated dynamics fixed and sweeping the contract space, the floor ratio $C_{\min}/C_{\max}$, the budget tightness $C_{\max}/(Nq_{\max})$, and the granularity $q_{\max}$, to establish which features of the policy are properties of the contract class rather than of the single contract studied here. Beyond both stands the assumption shared with the literature this thesis follows [3], that risk-neutral valuation applies to a volumetric claim, whereas the volume risk a physical holder carries is not obviously spanned. A hedging-based analysis of the same contract would test how much of the computed value survives incompleteness.

A Model Derivations

A.1 Equivalence to the Hull-White form

Differentiating $S_t = \alpha(t) + X_t$ of Definition 1 and substituting the dynamics of X_t from (2) gives

$$dS_t = \alpha'(t)\,dt + dX_t = \alpha'(t)\,dt - \kappa X_t\,dt + \sigma\,dW_t^{\mathbb{Q}}. \tag{61}$$

Substituting $X_t = S_t - \alpha(t)$ and collecting the drift,

$$dS_t = \kappa\Big(\alpha(t) + \frac{\alpha'(t)}{\kappa} - S_t\Big)dt + \sigma\,dW_t^{\mathbb{Q}}, \tag{62}$$

which is the classical form (3) with

$$\theta(t) = \alpha(t) + \frac{\alpha'(t)}{\kappa}. \tag{63}$$

For the inverse relation, read (63) as the linear first-order equation $\alpha'(t) = \kappa\big(\theta(t) - \alpha(t)\big)$ for α given θ. Its solution with initial value $\alpha(0)$ is

$$\alpha(t) = \alpha(0)\,e^{-\kappa t} + \kappa \int_0^t e^{-\kappa(t-u)}\,\theta(u)\,du, \tag{64}$$

verified by differentiation, with $\alpha(0) = S_0 - X_0$ fixed by the initial condition of Definition 1. Equations (63) and (64) map each specification to the other, so the two parameterisations carry the same information and the choice between them is one of convenience.

A.2 Solution of the deviation SDE and its conditional moments

The SDE for X_t in (2) is solved with the integrating factor $e^{\kappa t}$. Because $t \mapsto e^{\kappa t}$ is deterministic and of finite variation, the product rule for semimartingales applies with no cross-variation term, and

$$d\big(e^{\kappa t} X_t\big) = \kappa e^{\kappa t} X_t\,dt + e^{\kappa t}\,dX_t = \sigma e^{\kappa t}\,dW_t^{\mathbb{Q}}. \tag{65}$$

The drift term cancels exactly. Integrating (65) from s to t and multiplying by $e^{-\kappa t}$ gives

$$X_t = X_s e^{-\kappa(t-s)} + \sigma \int_s^t e^{-\kappa(t-u)}\,dW_u^{\mathbb{Q}}, \tag{66}$$

which is (5).

Conditioned on X_s, the first term is deterministic. The second is a Wiener integral of the deterministic, square-integrable integrand $u \mapsto e^{-\kappa(t-u)}$, hence a martingale increment with zero conditional mean, which gives (6). Its conditional variance follows from the Itô isometry,

$$\mathrm{Var}[X_t \mid X_s] = \sigma^2 \int_s^t e^{-2\kappa(t-u)}\,du = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(t-s)}\big), \tag{67}$$

which is (7). A Wiener integral of a deterministic integrand is Gaussian, so the transition law is

$$X_t \mid X_s \;\sim\; \mathcal{N}\bigg(X_s e^{-\kappa(t-s)},\; \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(t-s)}\big)\bigg), \tag{68}$$

the exact one-step law used in the estimation equation (50) and moment-matched by the lattice in Section 4.3.

A.3 The seasonal level under a constant market price of risk

Under the physical measure the decomposition reads $S_t = \alpha_{\mathbb{P}}(t) + X_t$ with $dX_t = -\kappa X_t\, dt + \sigma\, dW_t^{\mathbb{P}}$. A constant market price of risk λ changes measure through $dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \lambda\, dt$, as stated in Section 2.5. Substituting $dW_t^{\mathbb{P}} = dW_t^{\mathbb{Q}} - \lambda\, dt$ into the dynamics gives

$$dX_t = -\big(\kappa X_t + \sigma\lambda\big)\, dt + \sigma\, dW_t^{\mathbb{Q}}. \tag{69}$$

The added drift is a constant. Absorbing it into the reversion level, define the shifted deviation $\tilde{X}_t = X_t + \frac{\sigma\lambda}{\kappa}$. Then

$$d\tilde{X}_t = dX_t = -\kappa\Big(\tilde{X}_t - \frac{\sigma\lambda}{\kappa}\Big)dt - \sigma\lambda\, dt + \sigma\, dW_t^{\mathbb{Q}} = -\kappa\tilde{X}_t\, dt + \sigma\, dW_t^{\mathbb{Q}}. \tag{70}$$

so $\tilde{X}_t$ is a zero-mean Ornstein-Uhlenbeck process under $\mathbb{Q}$ with the same κ and σ. Rewriting the price in terms of the shifted deviation,

$$S_t = \alpha_{\mathbb{P}}(t) + X_t = \Big(\alpha_{\mathbb{P}}(t) - \frac{\sigma\lambda}{\kappa}\Big) + \tilde{X}_t, \tag{71}$$

identifies the risk-neutral seasonal level as $\alpha_{\mathbb{Q}}(t) = \alpha_{\mathbb{P}}(t) - \frac{\sigma\lambda}{\kappa}$, which is (9).

The constraint this places on the forward curve follows by taking $\mathbb{Q}$-expectations. Applying (8) to the $\mathbb{Q}$-decomposition and identifying the forward price with the expected spot through (47), the forward curve must satisfy, once the initial transient has decayed,

$$F(0,t) = \alpha_{\mathbb{P}}(t) - \frac{\sigma\lambda}{\kappa}. \tag{72}$$

The right side is the physical seasonal level translated by a single constant, so under a constant λ the forward curve must reproduce the seasonal shape of $\alpha_{\mathbb{P}}$ exactly, with unchanged amplitude and phase. Figure 1 shows the observed strip carries a stronger winter swing than the historical level, so no constant λ satisfies (72) at all t. The level is therefore fitted to the strip directly, $\alpha_{\mathbb{Q}}(t) = F(0,t)$ as in Section 5.2, while the invariance of κ and σ derived above is retained for the dynamics.

B Pricing and Control Derivations

B.1 Proof of the dynamic programming principle

Proof of Proposition 1. All quantities are taken on the lattice of Chapter 4. At each date the price takes one of the finitely many values $S_{i,j}$ of (17), so under any admissible strategy the pair (S_{t_i}, C_i) evolves as a finite-state Markov chain whose price transitions are the probabilities of Section 4.3, and every conditional expectation below is the finite sum (22).

Define $W_N(S,C)$ as the right side of (20) and, for $i = N-1, \ldots, 0$,

$$W_i(S,C) = \max_{q \in \mathcal{Q}_i(C)} \Big\{ q\, \xi(S) + e^{-r\Delta t}\, \mathbb{E}_{\mathbb{Q}}^{i,S,C}\big[W_{i+1}(S', C+q)\big] \Big\} \tag{73}$$

the right side of (21) with V_{i+1} replaced by W_{i+1}. Each feasible set $\mathcal{Q}_i(C)$ is finite and nonempty, since it contains $\underline{q}_i(C)$, so the maxima in (73) are attained. We prove by backward induction on i that, at every lattice state, $V_i = W_i$ and the value $V_i(S, C)$ is attained by an admissible strategy in $\mathcal{A}_i(C)$.

At $i = N$ a continuation strategy in $\mathcal{A}_N(C)$ consists of the single action $q_N \in \mathcal{Q}_N(C)$, and (19) reduces to the maximum of the immediate reward $q_N\, \xi(S)$ over that action, which is exactly (20). The maximising action attains it, establishing the base case.

Fix $i < N$ and a state (S, C), and assume the claim at date $i+1$. Let $\mathbf{q} = (q_i, \ldots, q_N) \in \mathcal{A}_i(C)$ be arbitrary. Splitting the sum in (19) at $k = i$ and writing $t_k - t_i = \Delta t + (t_k - t_{i+1})$ for $k > i$,

$$\mathbb{E}_{\mathbb{Q}}^{i,S,C}\left[\sum_{k=i}^{N} e^{-r(t_k - t_i)} q_k \xi(S_{t_k})\right] = \mathbb{E}_{\mathbb{Q}}^{i,S,C}\left[q_i \xi(S) + e^{-r\Delta t} \sum_{k=i+1}^{N} e^{-r(t_k - t_{i+1})} q_k \xi(S_{t_k})\right]. \quad (74)$$

Condition the inner sum on the information at t_{i+1} by the tower property. Given that information, the tail $(q_{i+1}, \ldots, q_N)$ is an admissible continuation strategy from the state $(i+1,\, S_{t_{i+1}},\, C + q_i)$, and by the Markov property of the chain the conditional law of the future prices depends on the past only through $S_{t_{i+1}}$. The conditional expectation of the tail reward is therefore at most the supremum over such continuations, which is $V_{i+1}(S_{t_{i+1}}, C + q_i) = W_{i+1}(S_{t_{i+1}}, C + q_i)$ by the inductive hypothesis. Substituting this bound into (74), the bracket becomes $q_i\, \xi(S) + e^{-r\Delta t}\, \mathbb{E}_{\mathbb{Q}}^{i,S,C}[W_{i+1}(S', C + q_i)]$, a function of the current state and the action q_i alone, and since q_i takes values in $\mathcal{Q}_i(C)$ this function is at most the maximum in (73). Hence the expected reward of every $\mathbf{q} \in \mathcal{A}_i(C)$ is at most $W_i(S, C)$, and taking the supremum over strategies gives $V_i(S, C) \le W_i(S, C)$.

For the reverse inequality, let q^* attain the maximum in (73). By the inductive hypothesis, for each successor state $(S', C + q^*)$ there is an admissible continuation strategy attaining $V_{i+1}(S', C + q^*)$. Define the strategy that takes q^* at (i, S, C) and, from date $i+1$ onward, follows the attaining continuation of whichever successor state is realised. Each of its actions is a function of the observed current state, so the strategy is adapted, and each action lies in its feasible set, so it belongs to $\mathcal{A}_i(C)$. Its expected reward is exactly

$$q^* \xi(S) + e^{-r\Delta t} \mathbb{E}_{\mathbb{Q}}^{i,S,C}\big[V_{i+1}(S', C + q^*)\big] = W_i(S, C), \quad (75)$$

so $V_i(S, C) \ge W_i(S, C)$ and the constructed strategy attains it. This closes the induction, so $V_i = W_i$ at every state, which is the terminal condition (20) together with the recursion (21).

Finally, $\mathcal{A}_0(0) = \mathcal{A}$ and $S_{t_0} = S_0$ is known, so evaluating the identity at the initial state $(0, S_0, 0)$ recovers (18), that is $V_0(S_0, 0) = V_0$. □

C Numerical Method Derivations

C.1 Transition probabilities at interior nodes

The system (33) to (35) determines the three interior probabilities. Dividing (34) by ΔX and (35) by ΔX^2, and writing $A = \frac{V + M^2}{2\Delta X^2}$ and $B = \frac{M}{2\Delta X}$ as in Section 4.3,

$$p_u^{(j)} - p_d^{(j)} = 2B, \qquad p_u^{(j)} + p_d^{(j)} = 2A. \quad (76)$$

Adding and subtracting the two equations gives $p_u^{(j)} = A + B$ and $p_d^{(j)} = A - B$, and substituting their sum into (33) gives $p_m^{(j)} = 1 - 2A$, which is (36). The conditions under which these probabilities lie in $[0, 1]$, and the role of the spacing (28) in guaranteeing them, are analysed in Section 4.3.

C.2 Transition probabilities at the boundary nodes

At the top node the three successors are the current node and the two below it, so the increment takes the values $\{0, -\Delta X, -2\Delta X\}$ with probabilities $p_u^{(j)}$, $p_m^{(j)}$, $p_d^{(j)}$. Matching the first moment M and the second raw moment $V + M^2$ of the increment, and dividing by ΔX and ΔX^2 as before,

$$p_m^{(j)} + 2\, p_d^{(j)} = -2B, \qquad p_m^{(j)} + 4\, p_d^{(j)} = 2A. \tag{77}$$

Subtracting the first equation from the second gives $p_d^{(j)} = A + B$, back-substitution gives $p_m^{(j)} = -2A - 4B$, and the sum condition gives $p_u^{(j)} = 1 - p_m^{(j)} - p_d^{(j)} = 1 + A + 3B$, which is (37).

At the bottom node the increments are $\{+2\Delta X, +\Delta X, 0\}$ for the branches $p_u^{(j)}$, $p_m^{(j)}$, $p_d^{(j)}$, and the same two conditions read

$$2\, p_u^{(j)} + p_m^{(j)} = 2B, \qquad 4\, p_u^{(j)} + p_m^{(j)} = 2A. \tag{78}$$

Subtracting gives $p_u^{(j)} = A - B$, back-substitution gives $p_m^{(j)} = -2A + 4B$, and the sum condition gives $p_d^{(j)} = 1 + A - 3B$, which is (38). In both cases the A and B terms cancel across the three branches and the residual 1 sits on the branch with zero increment, so each triple sums to one by construction.

C.3 Infeasibility of the binomial scheme under mean reversion

Consider a recombining binomial scheme for X_t on a grid of fixed spacing ΔX, in which the process moves from X_j to $X_j + \Delta X$ with probability $p^{(j)}$ and to $X_j - \Delta X$ with probability $1 - p^{(j)}$. The two branch probabilities sum to one, leaving a single free probability per node.

Matching the conditional mean of the increment, $M = -X_j\big(1 - e^{-\kappa \Delta t}\big)$ from (32), requires $\big(2p^{(j)} - 1\big)\Delta X = M$, hence

$$p^{(j)} = \frac{1}{2} + \frac{M}{2\Delta X}. \tag{79}$$

The second raw moment of the increment is $p^{(j)}\Delta X^2 + \big(1 - p^{(j)}\big)\Delta X^2 = \Delta X^2$, independent of the probability. The two-moment condition analogous to (35) therefore requires

$$\Delta X^2 = V + M^2. \tag{80}$$

The variance V is common to all nodes, but M^2 grows with X_j^2, so the right side of (80) is node-dependent while the spacing is fixed across the grid. The condition can hold at no more than one symmetric pair of nodes, those with $|X_j| = \sqrt{\Delta X^2 - V}\,/\big(1 - e^{-\kappa \Delta t}\big)$ when

$\Delta X^2 \geq V$, and fails at every other node. No recombining binomial grid with constant spacing can match both conditional moments of the Ornstein-Uhlenbeck increment. This obstruction is structural and holds at every time step and every parameter value with $\kappa > 0$.

Suppose the scheme instead abandons the variance and matches the mean alone, taking the spacing that carries the variance at the central node, $\Delta X = \sqrt{V}$. By (79), the probability lies in $[0, 1]$ if and only if $|M| \leq \Delta X$, that is

$$|X_j| \leq \frac{\sqrt{V}}{1 - e^{-\kappa \Delta t}}. \tag{81}$$

In units of the stationary standard deviation $\sigma/\sqrt{2\kappa}$ this radius is $\sqrt{1 - e^{-2\kappa\Delta t}}\,/\big(1 - e^{-\kappa\Delta t}\big)$, a decreasing function of $\kappa\Delta t$, so the stronger the per-step restoring drift, the further the infeasible region reaches into the grid. At the calibrated baseline ($\kappa = 29.2$, $\sigma = 12.75$, $\Delta t = 1/252$) the radius evaluates to 6.94 USD/MMBtu, or 4.16 stationary standard deviations. Rebuilding the grid of Section 4.2 at this spacing, $\sqrt{V} = 0.759$, the construction (31) gives $m_\sigma = 9$ and, with the buffer, $m = 11$, so the grid extends to $|X| = 8.35$. Its two outermost rows on each side, at $|X_j| = 7.59$ and 8.35, exceed the radius (81), and at $|X_j| = 7.59$ the mean-matching probability is $p = -0.047$. Even with the variance abandoned, the binomial probability leaves $[0, 1]$ on the grid the discretisation itself constructs, at nodes where the restoring drift exceeds what a single step can carry, and the infeasible region moves inward as $\kappa\Delta t$ grows. The trinomial system (33) to (35), with two free probabilities at the spacing (28), matches both moments at every node with the non-negativity margin analysed in Section 4.3.

D Validation Derivations

D.1 Closed-form value of the fully forced contract

Setting $C_{\min} = C_{\max} = (N{+}1)\, q_{\max}$ removes the local slack noted in Section 6.6, so the feasible set is the singleton $\mathcal{Q}_i(C) = \{q_{\max}\}$ at every date and the holder exercises the full cap unconditionally. No optimisation remains, and the value (19) at the initial state reduces to the deterministic discounted sum

$$V = \sum_{i=0}^{N} e^{-rt_i}\, q_{\max}\, \mathbb{E}_{\mathbb{Q}}\big[S_{t_i} - K\big]. \tag{82}$$

The expected spot is the conditional mean (8), $\mathbb{E}_{\mathbb{Q}}[S_{t_i}] = \alpha(t_i) + X_0 e^{-\kappa t_i}$, since the deviation has zero long-run mean and the seasonal level is deterministic. Substituting gives

$$V = \sum_{i=0}^{N} e^{-rt_i}\, q_{\max}\, \big(\alpha(t_i) + X_0 e^{-\kappa t_i} - K\big), \tag{83}$$

which is (60). Every term is a closed-form function of the calibrated parameters, so (83) carries no discretisation error and serves as an exact reference for the solver.

The comparison requires one adjustment. The lattice snaps the initial deviation to the nearest node before solving, so the dynamic program prices a contract that starts at the snapped deviation

$$X_0^{\text{snap}} = (j_0 - m)\,\Delta X_{\text{f}}, \qquad j_0 = \arg\min_j \big|X_j - X_0\big|, \tag{84}$$

rather than at X_0 itself. Evaluating (83) at X_0^{snap} in place of X_0 matches the initial condition the solver actually uses, which isolates discretisation error from a mismatch in the starting point. Against this target the solver agrees to within 0.02% at every resolution level, reported per resolution level in Table 5.

E Supplementary Figures and Tables

The entire convergence ladder is presented in table 5.

Table 5: Full convergence ladder for K_{sub}

K_{sub}	N_{fine}	ΔX_f	Price	Δ Price	Clips	$\bar{S}^*\big\|_{C=0}$	Memory (GB)		Runtime (s)
							Pred.	Peak	
1	252	1.3911	1572.9840	—	0	2.6404	0.019	56.43	0.39
4	1008	0.6956	1583.4231	+10.4391	0	2.6562	0.127	56.43	0.58
8	2016	0.4918	1583.8661	+0.4430	0	2.6604	0.336	56.43	0.89
16	4032	0.3478	1571.7238	−12.1423	2	2.6623	0.915	56.43	1.69
32	8064	0.2459	1575.0450	+3.3213	2	2.6632	2.481	56.43	3.45
64	16128	0.1739	1577.4167	+2.3717	2	2.6635	6.752	56.43	7.23
100	25200	0.1391	1573.7374	−3.6793	2	2.6635	12.838	56.43	13.04
128	32256	0.1230	1574.8558	+1.1184	2	2.6633	18.711	56.43	66.69
256	64512	0.0869	1574.2767	−0.5791	2	2.6635	51.739	57.64	157.95

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